

BSDS Exercise Set 1

Detailed step-by-step solutions in L^AT_EX

1. Probability generating functions

Full question. Let P, P_1 and P_2 be pgfs.

- a) Show that $\lambda P_1 + (1 - \lambda)P_2$ is a pgf for any $0 \leq \lambda \leq 1$. Show that $P_1 P_2$ is a pgf.
- b) Is $\exp(P(s) - 1)$ a pgf? Fix $0 < \lambda < 1$. Is $P(\lambda s)/P(\lambda)$ necessarily a pgf?

Theory used. A pgf is a power series with nonnegative coefficients summing to 1. The book also notes that a pgf is nondecreasing and convex on $[0, 1]$, and that products of pgfs correspond to sums of independent random variables.

Solution.

Let

$$P_1(s) = \sum_{k \geq 0} a_k s^k, \quad P_2(s) = \sum_{k \geq 0} b_k s^k,$$

where $a_k, b_k \geq 0$ and $\sum_{k \geq 0} a_k = \sum_{k \geq 0} b_k = 1$.

(a) Convex combination. For $0 \leq \lambda \leq 1$,

$$\lambda P_1(s) + (1 - \lambda)P_2(s) = \sum_{k \geq 0} (\lambda a_k + (1 - \lambda)b_k) s^k.$$

Each coefficient is nonnegative, and

$$\sum_{k \geq 0} (\lambda a_k + (1 - \lambda)b_k) = \lambda \sum_{k \geq 0} a_k + (1 - \lambda) \sum_{k \geq 0} b_k = \lambda + (1 - \lambda) = 1.$$

Hence $\lambda P_1 + (1 - \lambda)P_2$ is a pgf.

Product. Now

$$P_1(s)P_2(s) = \left(\sum_{k \geq 0} a_k s^k \right) \left(\sum_{m \geq 0} b_m s^m \right) = \sum_{n \geq 0} c_n s^n,$$

where

$$c_n = \sum_{k=0}^n a_k b_{n-k} \geq 0$$

by the Cauchy product. Also,

$$\sum_{n \geq 0} c_n = \left(\sum_{k \geq 0} a_k \right) \left(\sum_{m \geq 0} b_m \right) = 1.$$

Therefore $P_1 P_2$ is a pgf.

(b) The function $\exp(P(s) - 1)$. Since $P(s)$ is a pgf, we may write

$$P(s) = \sum_{k \geq 0} p_k s^k, \quad p_k \geq 0, \quad \sum_{k \geq 0} p_k = 1.$$

Then

$$e^{P(s)-1} = e^{-1} \exp(P(s)) = e^{-1} \sum_{n=0}^{\infty} \frac{P(s)^n}{n!}.$$

For each fixed n , $P(s)^n$ is a pgf because it is the product of n pgfs. The weights $e^{-1}/n!$ are nonnegative and sum to 1:

$$\sum_{n=0}^{\infty} \frac{e^{-1}}{n!} = 1.$$

So $\exp(P(s) - 1)$ is a convex combination of pgfs, hence itself a pgf.

The tilted function $\frac{P(\lambda s)}{P(\lambda)}$. Write

$$\frac{P(\lambda s)}{P(\lambda)} = \frac{\sum_{k \geq 0} p_k (\lambda s)^k}{\sum_{k \geq 0} p_k \lambda^k} = \sum_{k \geq 0} \left(\frac{p_k \lambda^k}{P(\lambda)} \right) s^k.$$

All coefficients are nonnegative. Their sum is

$$\sum_{k \geq 0} \frac{p_k \lambda^k}{P(\lambda)} = \frac{P(\lambda)}{P(\lambda)} = 1.$$

Hence, for every $0 < \lambda < 1$, the function $P(\lambda s)/P(\lambda)$ is also a pgf.

2. Finding pgfs and radii of convergence

Full question. Find the pgf for the following random variables and state their radius of convergence:

- a) X where $\mathbb{P}(X = n) = \frac{c}{n!}$ for $n \geq 0$. Find the constant c first.
- b) Y where $\mathbb{P}(Y = n) = \frac{c}{(n+1)(n+2)}$ for $n \geq 0$. Find the constant c first.
- c) Z where $\mathbb{P}(Z = n) = \frac{1}{2^{n+1}}$ for $n \geq 0$.

Theory used. The book states that a pgf is a power series, is uniquely determined by its coefficients, and has radius of convergence at least 1 for a nonnegative integer-valued random variable.

Solution.

(a)

$$\sum_{n \geq 0} \frac{c}{n!} = c \sum_{n \geq 0} \frac{1}{n!} = ce = 1 \implies c = \frac{1}{e}.$$

Hence

$$P_X(s) = \sum_{n \geq 0} \frac{1}{e} \frac{s^n}{n!} = \frac{1}{e} e^s = e^{s-1}.$$

This is an entire function, so the radius of convergence is ∞ .

(b) First determine c :

$$\sum_{n \geq 0} \frac{c}{(n+1)(n+2)} = c \sum_{n \geq 0} \left(\frac{1}{n+1} - \frac{1}{n+2} \right) = c.$$

Therefore $c = 1$.

Now

$$P_Y(s) = \sum_{n \geq 0} \frac{s^n}{(n+1)(n+2)} = \sum_{n \geq 0} \left(\frac{s^n}{n+1} - \frac{s^n}{n+2} \right).$$

Use the classical series

$$\sum_{n \geq 0} \frac{s^n}{n+1} = -\frac{\ln(1-s)}{s}, \quad \sum_{n \geq 0} \frac{s^n}{n+2} = \frac{-\ln(1-s) - s}{s^2}.$$

Therefore

$$P_Y(s) = -\frac{\ln(1-s)}{s} - \frac{-\ln(1-s) - s}{s^2} = \frac{s + (1-s)\ln(1-s)}{s^2}.$$

The nearest singularity is at $s = 1$, so the radius of convergence is 1.

(c)

$$P_Z(s) = \sum_{n \geq 0} \frac{s^n}{2^{n+1}} = \frac{1}{2} \sum_{n \geq 0} \left(\frac{s}{2} \right)^n = \frac{1}{2-s}.$$

The radius of convergence is 2.

3. Distribution from a pgf

Full question. Find the distribution of the random variables in each of the cases and find $\mathbb{P}(X \geq n)$ for $n \geq 1$.

a) $P(s) = \frac{1}{3-2s}.$

b) $P(s) = \frac{1}{4-3s}.$

c) Two dice are rolled. Let X be the sum of the outcomes. Find the pgf of X . Use this pgf to calculate the probability that the sum is divisible by 3.

d) If the pgf of a random variable X is

$$P(s) = \frac{1}{2}e^{s-1} + \frac{1}{4}s^2 + \frac{1}{4},$$

find $\mathbb{P}(X \geq 2)$.

Theory used. The book's pgf examples include geometric distributions, tail probabilities via pgfs, and the product rule for independent sums. The relation $\mathbb{P}(X \geq n) = 1 - \mathbb{P}(X < n)$ is also used.

Solution.

(a)

$$P(s) = \frac{1}{3-2s} = \frac{1}{3} \cdot \frac{1}{1-\frac{2}{3}s} = \sum_{k \geq 0} \frac{1}{3} \left(\frac{2}{3}\right)^k s^k.$$

So

$$\mathbb{P}(X = k) = \frac{1}{3} \left(\frac{2}{3}\right)^k, \quad k \geq 0,$$

which is a geometric distribution on $\{0, 1, 2, \dots\}$ with parameter $p = \frac{1}{3}$.

Therefore,

$$\mathbb{P}(X \geq n) = \sum_{k=n}^{\infty} \frac{1}{3} \left(\frac{2}{3}\right)^k = \left(\frac{2}{3}\right)^n, \quad n \geq 1.$$

(b)

$$P(s) = \frac{1}{4-3s} = \frac{1}{4} \cdot \frac{1}{1-\frac{3}{4}s} = \sum_{k \geq 0} \frac{1}{4} \left(\frac{3}{4}\right)^k s^k.$$

So

$$\mathbb{P}(X = k) = \frac{1}{4} \left(\frac{3}{4}\right)^k, \quad k \geq 0,$$

a geometric distribution on $\{0, 1, 2, \dots\}$ with parameter $p = \frac{1}{4}$.

Hence

$$\mathbb{P}(X \geq n) = \left(\frac{3}{4}\right)^n, \quad n \geq 1.$$

(c) Let U be the outcome of one fair die. Then

$$P_U(s) = \frac{s + s^2 + s^3 + s^4 + s^5 + s^6}{6}.$$

Since the two dice are independent, the pgf of their sum is

$$P_X(s) = \left(\frac{s + s^2 + s^3 + s^4 + s^5 + s^6}{6} \right)^2.$$

To find the probability that the sum is divisible by 3, use the roots of unity filter. Let

$$\omega = e^{2\pi i/3}, \quad \omega^2 + \omega + 1 = 0.$$

Then

$$\mathbb{P}(X \equiv 0 \pmod{3}) = \frac{1}{3} \left(P_X(1) + P_X(\omega) + P_X(\omega^2) \right).$$

Now

$$1 + \omega + \omega^2 = 0 \implies 1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 = 0,$$

so $P_U(\omega) = P_U(\omega^2) = 0$. Therefore

$$P_X(\omega) = P_X(\omega^2) = 0, \quad P_X(1) = 1,$$

and hence

$$\mathbb{P}(X \equiv 0 \pmod{3}) = \frac{1}{3}.$$

(d) Read the expression as

$$P(s) = \frac{1}{2}e^{s-1} + \frac{1}{4}s^2 + \frac{1}{4},$$

which is the only pgf-consistent interpretation.

Now

$$P(0) = \frac{1}{2}e^{-1} + \frac{1}{4}, \quad P'(s) = \frac{1}{2}e^{s-1} + \frac{1}{2}s,$$

so

$$\mathbb{P}(X = 1) = P'(0) = \frac{1}{2}e^{-1}.$$

Therefore

$$\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1) = 1 - \left(\frac{1}{2}e^{-1} + \frac{1}{4}\right) - \frac{1}{2}e^{-1} = \frac{3}{4} - \frac{1}{e}.$$

4. Generating functions and convolution

Full question.

a) Use generating functions to prove the combinatorial identity

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$$

b) Find the sequence whose generating function is $\frac{1}{(1-s)^4}$ by expressing it as a product of known generating functions and using convolution.

c) Let a_n be the number of ways to tile a $2 \times n$ board with 1×2 dominoes for $n \geq 1$. Find a recurrence relation for a_n for $n \geq 3$. Set a_1, a_2 appropriately and find its generating function.

Theory used. The book states: “the generating function of $X + Y$ is given by product of the generating function of X and Y ” and introduces convolution of sequences.

Solution.

(a) Consider

$$(1+s)^n(1+s)^n = (1+s)^{2n}.$$

The coefficient of s^n on the left is

$$\sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \sum_{k=0}^n \binom{n}{k}^2.$$

The coefficient of s^n on the right is $\binom{2n}{n}$. Therefore,

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$$

(b) We know

$$\frac{1}{(1-s)^2} = \sum_{n \geq 0} (n+1)s^n.$$

Hence

$$\frac{1}{(1-s)^4} = \frac{1}{(1-s)^2} \cdot \frac{1}{(1-s)^2}.$$

By convolution, the coefficient of s^n is

$$c_n = \sum_{k=0}^n (k+1)(n-k+1).$$

Now compute:

$$c_n = \sum_{k=0}^n ((k+1)(n+1-k)) = \frac{(n+1)(n+2)(n+3)}{6} = \binom{n+3}{3}.$$

So

$$\frac{1}{(1-s)^4} = \sum_{n \geq 0} \binom{n+3}{3} s^n.$$

Thus the sequence is $\left\{ \binom{n+3}{3} \right\}_{n \geq 0}$.

(c) A $2 \times n$ board can be tiled in two ways according to the first part:

- place one vertical pair of dominoes in the first column, leaving a $2 \times (n-1)$ board;
- place two horizontal dominoes in the first two columns, leaving a $2 \times (n-2)$ board.

Hence for $n \geq 3$,

$$a_n = a_{n-1} + a_{n-2}.$$

The initial values are

$$a_1 = 1, \quad a_2 = 2.$$

Let

$$A(s) = \sum_{n \geq 1} a_n s^n.$$

Then

$$A(s) = s + 2s^2 + \sum_{n \geq 3} (a_{n-1} + a_{n-2}) s^n.$$

Now

$$\sum_{n \geq 3} a_{n-1} s^n = s \sum_{m \geq 2} a_m s^m = s(A(s) - s),$$

and

$$\sum_{n \geq 3} a_{n-2} s^n = s^2 \sum_{m \geq 1} a_m s^m = s^2 A(s).$$

Therefore

$$A(s) = s + 2s^2 + s(A(s) - s) + s^2 A(s).$$

Rearranging,

$$A(s)(1 - s - s^2) = s + s^2,$$

so

$$A(s) = \frac{s + s^2}{1 - s - s^2}.$$

5. First occurrence of patterns and residues modulo 3

Full question.

- a) Let p_k be the probability that three consecutive Heads (HHH) is obtained for the first time after the k -th toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.
- b) Let p_k be the probability of the pattern of two consecutive Heads (HH) or the pattern of Head followed by Tail (HT) is obtained for the first time after the k -th toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.
- c) Let p_k be the probability that the total obtained in k throws of a dice (not necessarily fair) is divisible by 3. Find the generating function of $\{p_k : k \geq 0\}$. Find $\lim_{k \rightarrow \infty} p_k$.

Theory used. The book's generating-function chapter uses first-step recursions and tail/mean identities, and the later chapters use Markov-chain hitting-time recursions. The key idea here is to set up a small state recursion for the waiting time.

Solution.

(a) **Three consecutive heads.** Let T be the first time the pattern HHH appears. Define

$$F_i(s) = \mathbb{E}[s^T \mid \text{current run of consecutive heads is } i], \quad i = 0, 1, 2.$$

We need $F_0(s)$.

From state 0:

$$F_0(s) = \frac{s}{2}F_0(s) + \frac{s}{2}F_1(s).$$

From state 1:

$$F_1(s) = \frac{s}{2}F_0(s) + \frac{s}{2}F_2(s).$$

From state 2:

$$F_2(s) = \frac{s}{2}F_0(s) + \frac{s}{2}.$$

The last equation is because a head completes HHH in one more toss.

Now solve:

$$F_2(s) = \frac{s}{2}(1 + F_0(s)),$$

$$F_1(s) = \frac{s}{2}F_0(s) + \frac{s}{2}F_2(s) = \frac{s}{2}F_0(s) + \frac{s^2}{4}(1 + F_0(s)),$$

and then

$$F_0(s) = \frac{s}{2}F_0(s) + \frac{s}{2}F_1(s).$$

Substituting and simplifying gives

$$F_0(s) = \frac{s^3}{8 - 4s - 2s^2 - s^3}.$$

Hence the generating function of $\{p_k\}$ is

$$\sum_{k \geq 0} p_k s^k = \frac{s^3}{8 - 4s - 2s^2 - s^3}.$$

The expected waiting time is

$$\mathbb{E}[T] = F'_0(1) = 14.$$

(b) Pattern HH or HT. The first time either HH or HT appears is exactly one step after the first Head occurs. Indeed, once the first Head is observed, the very next toss completes either HH or HT.

Let N be the time of the first Head. For a fair coin,

$$\mathbb{P}(N = n) = 2^{-n}, \quad n \geq 1.$$

If T denotes the required time, then $T = N + 1$, so

$$\mathbb{P}(T = k) = 2^{-(k-1)}, \quad k \geq 2.$$

Therefore

$$\sum_{k \geq 0} p_k s^k = \sum_{k \geq 2} 2^{-(k-1)} s^k = \frac{s^2}{2-s}.$$

The expected time is

$$\mathbb{E}[T] = \sum_{k \geq 2} k 2^{-(k-1)} = 3.$$

(c) Sum divisible by 3. Let the probabilities of the die outcomes be $r_1, \dots, r_6$ and define the residue-class probabilities

$$u_0 = r_3 + r_6, \quad u_1 = r_1 + r_4, \quad u_2 = r_2 + r_5.$$

Let $\omega = e^{2\pi i/3}$.

For one throw, the root-of-unity filter gives

$$\alpha := u_0 + u_1\omega + u_2\omega^2, \quad \bar{\alpha} = u_0 + u_1\omega^2 + u_2\omega.$$

Then for the sum of k independent throws,

$$p_k = \mathbb{P}(S_k \equiv 0 \pmod{3}) = \frac{1}{3}(1 + \alpha^k + \bar{\alpha}^k).$$

Therefore the generating function is

$$\sum_{k \geq 0} p_k s^k = \frac{1}{3} \left(\frac{1}{1-s} + \frac{1}{1-\alpha s} + \frac{1}{1-\bar{\alpha} s} \right).$$

For the limit, if the die is not concentrated on a single residue class mod 3, then $|\alpha| < 1$, so

$$\lim_{k \rightarrow \infty} p_k = \frac{1}{3}.$$

In the degenerate case where all mass lies in one residue class, the sequence is periodic rather than convergent.

6. Random sums and Poisson thinning

Full question.

- Claims to an insurance company arrive as a Poisson process with rate λ . The size of each claim is an iid random variable Y with pgf $P_Y(s)$. Let S_n be the total claim amount up to time n . Find the pgf of S_n .
- A hen lays N eggs, where $N \sim \text{Poisson}(\lambda)$. Each egg hatches with probability p independently. Let K be the number of chicks. Find the pgf of K and identify its distribution.
- In the previous problem, find the expected number of chicks, $\mathbb{E}[K]$, using the random sum formula and verify by direct computation.

Theory used. The book states: “the pgf of S_N is given by $G(P(s))$ ” for an independent random sum, and also that the mean of a random sum satisfies $\mathbb{E}(S_N) = \mathbb{E}(N)\mathbb{E}(X_1)$.

Solution.

(a) Let N_n be the number of claims by time n . Since claims arrive according to a Poisson process with rate λ ,

$$N_n \sim \text{Poisson}(\lambda n).$$

If the claim sizes are $Y_1, Y_2, \dots$ with common pgf $P_Y(s)$ and

$$S_n = \sum_{i=1}^{N_n} Y_i,$$

then by the random-sum formula,

$$P_{S_n}(s) = \mathbb{E}[P_Y(s)^{N_n}].$$

Because $N_n \sim \text{Poisson}(\lambda n)$, its pgf is

$$G_{N_n}(u) = \exp(\lambda n(u - 1)).$$

Hence

$$P_{S_n}(s) = \exp(\lambda n(P_Y(s) - 1)).$$

(b) Conditional on N , the number of chicks is

$$K \mid N \sim \text{Bin}(N, p).$$

Therefore

$$\mathbb{E}[s^K \mid N] = (1 - p + ps)^N.$$

Taking expectation over $N \sim \text{Poisson}(\lambda)$,

$$P_K(s) = \mathbb{E}[(1 - p + ps)^N] = \exp(\lambda((1 - p + ps) - 1)) = \exp(\lambda p(s - 1)).$$

This is exactly the pgf of a $\text{Poisson}(\lambda p)$ random variable. So

$$K \sim \text{Poisson}(\lambda p).$$

(c) Using the random-sum mean formula,

$$\mathbb{E}[K] = \mathbb{E}[N]\mathbb{E}[\text{chicks from one egg}] = \lambda \cdot p = \lambda p.$$

Directly,

$$\mathbb{E}[K] = \mathbb{E}(\mathbb{E}[K \mid N]).$$

Given $N = n$, we have $\mathbb{E}[K \mid N = n] = np$, so

$$\mathbb{E}[K] = \sum_{n \geq 0} np e^{-\lambda} \frac{\lambda^n}{n!} = p e^{-\lambda} \sum_{n \geq 1} n \frac{\lambda^n}{n!}.$$

Now

$$n \frac{\lambda^n}{n!} = \lambda \frac{\lambda^{n-1}}{(n-1)!},$$

hence

$$\mathbb{E}[K] = p e^{-\lambda} \lambda \sum_{m \geq 0} \frac{\lambda^m}{m!} = p e^{-\lambda} \lambda e^{\lambda} = \lambda p.$$

This agrees with the random-sum formula.

BSDS Exercise Set 2

Detailed step-by-step L^AT_EX solutions

Note on the OCR text in Question 1(b). The uploaded PDF repeats the symbol $P(\xi = 2)$ twice. In the solution below, I interpret this as the intended offspring distribution

$$\mathbb{P}(\xi = 0) = \frac{1}{6}, \quad \mathbb{P}(\xi = 1) = \frac{1}{2}, \quad \mathbb{P}(\xi = 2) = \frac{1}{6}, \quad \mathbb{P}(\xi = 3) = \frac{1}{6},$$

so that the probabilities sum to 1. The method is the same for any corrected version of the typo.

Question

Problem 1. Consider a branching process starting with a single individual, $Z_0 = 1$.

- (a) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = \frac{1}{4}$, $\mathbb{P}(\xi = 1) = \frac{1}{2}$, $\mathbb{P}(\xi = 2) = \frac{1}{4}$, compute $\mathbb{E}(Z_3)$ and $\text{Var}(Z_3)$. Also, find the probability that $\mathbb{P}(Z_3 = 0)$.
- (b) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = \frac{1}{6}$, $\mathbb{P}(\xi = 1) = \frac{1}{2}$, $\mathbb{P}(\xi = 2) = \frac{1}{6}$, $\mathbb{P}(\xi = 3) = \frac{1}{6}$, compute $\mathbb{E}(Z_2 Z_3)$ and $\text{Var}(Z_2 + Z_3)$. Also, find the probability $\mathbb{P}(Z_2 > 0, Z_3 = 0)$.

Theory used

For a Galton–Watson branching process with offspring pgf $G(s) = \mathbb{E}(s^\xi)$, the pgf of the n -th generation is the n -fold iterate $G_n = G \circ G \circ \cdots \circ G$. Also,

$$\mathbb{E}(Z_n) = \mu^n, \quad \mu = \mathbb{E}(\xi),$$

and if $\sigma^2 = \text{Var}(\xi) < \infty$, then

$$\text{Var}(Z_n) = \begin{cases} n\sigma^2, & \mu = 1, \\ \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1}, & \mu \neq 1. \end{cases}$$

Moreover, $\mathbb{P}(Z_n = 0) = G_n(0)$.

(a) Step 1: Write the offspring pgf.

The offspring distribution is

$$G(s) = \frac{1}{4} + \frac{1}{2}s + \frac{1}{4}s^2.$$

It is convenient to factor it:

$$G(s) = \frac{(1+s)^2}{4}.$$

Step 2: Compute the mean and variance of the offspring distribution.

First,

$$\mu = \mathbb{E}(\xi) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1.$$

Next,

$$\mathbb{E}(\xi^2) = 0^2 \cdot \frac{1}{4} + 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4} = \frac{1}{2} + 1 = \frac{3}{2}.$$

Hence

$$\sigma^2 = \text{Var}(\xi) = \mathbb{E}(\xi^2) - \mu^2 = \frac{3}{2} - 1 = \frac{1}{2}.$$

Step 3: Compute $\mathbb{E}(Z_3)$.

For a branching process,

$$\mathbb{E}(Z_n) = \mu^n.$$

Since $\mu = 1$,

$$\mathbb{E}(Z_3) = 1^3 = 1.$$

Step 4: Compute $\text{Var}(Z_3)$.

Because $\mu = 1$, the variance formula becomes

$$\text{Var}(Z_n) = n\sigma^2.$$

Therefore,

$$\text{Var}(Z_3) = 3 \cdot \frac{1}{2} = \frac{3}{2}.$$

Step 5: Compute $\mathbb{P}(Z_3 = 0)$.

The extinction probability at time 3 is

$$\mathbb{P}(Z_3 = 0) = G_3(0) = G(G(G(0))).$$

Now compute this step by step:

$$G(0) = \frac{1}{4},$$

$$G_2(0) = G\left(\frac{1}{4}\right) = \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{16} = \frac{1}{4} + \frac{1}{8} + \frac{1}{64} = \frac{25}{64},$$

and then

$$G_3(0) = G\left(\frac{25}{64}\right) = \frac{1}{4} + \frac{1}{2} \cdot \frac{25}{64} + \frac{1}{4} \cdot \left(\frac{25}{64}\right)^2.$$

Putting everything over a common denominator,

$$G_3(0) = \frac{1}{4} + \frac{25}{128} + \frac{625}{16384} = \frac{4096 + 3200 + 625}{16384} = \frac{7921}{16384}.$$

Theory used

For branching processes, conditioning on the present generation gives

$$\mathbb{E}(Z_{n+1} \mid Z_n) = \mu Z_n.$$

This implies

$$\mathbb{E}(Z_n Z_m) = \mu^{n-m} \mathbb{E}(Z_m^2), \quad m \leq n,$$

and therefore

$$\text{Cov}(Z_m, Z_n) = \mu^{n-m} \text{Var}(Z_m).$$

Also, if $n > m$, then

$$\mathbb{E}(Z_n \mid Z_m = k) = k\mu^{n-m}.$$

(b) Step 1: Identify the offspring pgf.

With the corrected interpretation,

$$G(s) = \frac{1}{6} + \frac{1}{2}s + \frac{1}{6}s^2 + \frac{1}{6}s^3.$$

Step 2: Compute the mean and variance of one offspring.

We calculate

$$\mu = \mathbb{E}(\xi) = 0 \cdot \frac{1}{6} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} = \frac{1}{2} + \frac{1}{3} + \frac{1}{2} = \frac{4}{3}.$$

Also,

$$\mathbb{E}(\xi^2) = 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{6} + 3^2 \cdot \frac{1}{6} = \frac{1}{2} + \frac{4}{6} + \frac{9}{6} = \frac{8}{3}.$$

Hence

$$\sigma^2 = \text{Var}(\xi) = \frac{8}{3} - \left(\frac{4}{3}\right)^2 = \frac{8}{3} - \frac{16}{9} = \frac{8}{9}.$$

Step 3: Compute $\mathbb{E}(Z_2 Z_3)$.

Use

$$\mathbb{E}(Z_3 \mid Z_2) = \mu Z_2.$$

So

$$\mathbb{E}(Z_2 Z_3) = \mathbb{E}(Z_2 \mathbb{E}(Z_3 \mid Z_2)) = \mathbb{E}(Z_2 \cdot \mu Z_2) = \mu \mathbb{E}(Z_2^2).$$

Now

$$\mathbb{E}(Z_2) = \mu^2 = \left(\frac{4}{3}\right)^2 = \frac{16}{9}.$$

The variance at generation 2 is

$$\text{Var}(Z_2) = \sigma^2 \mu \frac{\mu^2 - 1}{\mu - 1} = \frac{8}{9} \cdot \frac{4}{3} \cdot \frac{\frac{16}{9} - 1}{\frac{4}{3} - 1} = \frac{224}{81}.$$

Hence

$$\mathbb{E}(Z_2^2) = \text{Var}(Z_2) + \mathbb{E}(Z_2)^2 = \frac{224}{81} + \left(\frac{16}{9}\right)^2 = \frac{224}{81} + \frac{256}{81} = \frac{160}{27}.$$

Therefore,

$$\mathbb{E}(Z_2 Z_3) = \frac{4}{3} \cdot \frac{160}{27} = \frac{640}{81}.$$

Step 4: Compute $\text{Var}(Z_2 + Z_3)$.

We use

$$\text{Var}(Z_2 + Z_3) = \text{Var}(Z_2) + \text{Var}(Z_3) + 2\text{Cov}(Z_2, Z_3).$$

We already have $\text{Var}(Z_2) = 224/81$. Also,

$$\text{Var}(Z_3) = \sigma^2 \mu^2 \frac{\mu^3 - 1}{\mu - 1} = \frac{8}{9} \cdot \left(\frac{4}{3}\right)^2 \cdot \frac{\left(\frac{4}{3}\right)^3 - 1}{\frac{4}{3} - 1} = \frac{4736}{729}.$$

Next,

$$\text{Cov}(Z_2, Z_3) = \mu^{3-2} \text{Var}(Z_2) - \mu \text{Var}(Z_2) = \frac{4}{3} \cdot \frac{224}{81} = \frac{896}{243}.$$

Thus

$$\text{Var}(Z_2 + Z_3) = \frac{224}{81} + \frac{4736}{729} + 2 \cdot \frac{896}{243} = \frac{12128}{243}.$$

Question

Problem 2. Suppose that $\{Z_n : n \geq 0\}$ is a branching process with $Z_0 = 1$. Fix any positive integer $r \geq 1$ and define $\{Y_n : n \geq 0\}$ by

$$Y_n = Z_{nr}, \quad n \geq 0.$$

Check that $\{Y_n\}$ is also a branching process. Find its probability generating function of the offspring distribution.

Theory used

If G is the offspring pgf of a branching process, then the pgf of the n -th generation is the n -fold iterate $G_n = G^{\circ n}$. In particular, the evolution over r generations acts like one new generation whose offspring pgf is G_r .

Step-by-step solution

Step 1: Interpret Y_n as a coarse-grained version of Z_n .

The variable $Y_n = Z_{nr}$ records the population every r -th generation only. So one step of $\{Y_n\}$ corresponds to r steps of the original branching process.

Step 2: Use the branching property.

Starting from one individual, the number of individuals after r generations has pgf

$$G_r(s) = \underbrace{G(G(\cdots G(s)\cdots))}_{r \text{ times}}.$$

Thus, if one individual of the Y -process produces the population after r generations, then its offspring pgf is exactly G_r .

Step 3: Verify the branching structure.

If $Y_n = k$, then there are k independent families at time nr , and each family evolves independently for the next r generations. Hence the next state Y_{n+1} is the sum of k independent copies of the r -generation offspring count. That is precisely the branching property.

Conclusion.

Therefore $\{Y_n\}$ is also a branching process, and its offspring pgf is

$$G_Y(s) = G_r(s) = G^{\circ r}(s).$$

Question

Problem 3. Consider a branching process with offspring pgf

$$G(s) = \frac{q}{1 - ps}, \quad 0 < p < 1, \quad q = 1 - p.$$

(a) For $p = \frac{1}{2}$, show that

$$G_n(s) = \frac{n - (n-1)s}{n + 1 - ns}, \quad n \geq 1,$$

and show that $G_n(0) \rightarrow 1$ as $n \rightarrow \infty$.

(b) For $p \neq \frac{1}{2}$, derive an explicit formula for $G_n(s)$. Further conclude that

$$\lim_{n \rightarrow \infty} G_n(0) = \begin{cases} 1, & p < q, \\ \frac{q}{p}, & p > q. \end{cases}$$

Theory used

For a branching process, the pgf at generation n is the n -fold iterate $G_n = G^{\circ n}$. Also, the extinction probability is the smallest fixed point of $G(s) = s$ in $[0, 1]$. When G is linear fractional, iterates can be found by induction or by a change of variables that turns composition into multiplication.

Step-by-step solution

(a) Step 1: Specialize the pgf.

If $p = q = \frac{1}{2}$, then

$$G(s) = \frac{1/2}{1 - \frac{1}{2}s} = \frac{1}{2 - s}.$$

Step 2: Guess the form and verify by induction.

We claim

$$G_n(s) = \frac{n - (n-1)s}{n+1 - ns}, \quad n \geq 1.$$

For $n = 1$,

$$G_1(s) = G(s) = \frac{1}{2 - s},$$

and this is exactly

$$\frac{1 - (1-1)s}{1 + 1 - 1 \cdot s} = \frac{1}{2 - s}.$$

Assume the formula holds for n . Then

$$G_{n+1}(s) = G(G_n(s)) = \frac{1}{2 - G_n(s)}.$$

Substitute the inductive hypothesis:

$$G_{n+1}(s) = \frac{1}{2 - \frac{n - (n-1)s}{n+1 - ns}}.$$

Put over a common denominator:

$$2 - \frac{n - (n-1)s}{n+1 - ns} = \frac{2(n+1 - ns) - (n - (n-1)s)}{n+1 - ns} = \frac{n+2 - (n+1)s}{n+1 - ns}.$$

Therefore

$$G_{n+1}(s) = \frac{n+1 - ns}{n+2 - (n+1)s} = \frac{(n+1) - ns}{(n+1) + 1 - (n+1)s},$$

which is the same formula with n replaced by $n+1$. Hence the claim is proved.

Step 3: Evaluate $G_n(0)$.

Setting $s = 0$,

$$G_n(0) = \frac{n}{n+1} \xrightarrow{n \rightarrow \infty} 1.$$

Conclusion for (a):

$$G_n(s) = \frac{n - (n-1)s}{n+1 - ns}, \quad G_n(0) \rightarrow 1.$$

(b) Step 1: Find a useful fixed-point transform.

Let

$$r = \frac{q}{p}.$$

Then r is the nontrivial fixed point of the map $G(s)$, because

$$s = \frac{q}{1 - ps} \iff ps^2 - s + q = 0 \iff (s-1) \left(s - \frac{q}{p} \right) = 0.$$

Question

Problem 4. Let $\{Z_n\}$ be a branching process with $Z_0 = 1$ and offspring mean μ .

(a) Assume that $G''(1) < \infty$. Show that

$$\mathbb{E}(Z_n Z_m) = \mu^{n-m} \mathbb{E}(Z_m^2), \quad m \leq n.$$

Hence find the covariance and correlation coefficient between Z_m and Z_n (in terms of μ) for $m \leq n$.

(b) Find $\mathbb{E}(Z_n \mid Z_m = k)$ for $m < n$ and $k \geq 0$.

Theory used

For a branching process, the conditional mean is

$$\mathbb{E}(Z_n \mid Z_m) = \mu^{n-m} Z_m, \quad n \geq m.$$

Also, if $G''(1) < \infty$, then $\text{Var}(Z_n)$ is finite and can be written in terms of μ and $\sigma^2 = \text{Var}(\xi)$. The law of total expectation and the law of total variance are the main tools.

Step-by-step solution

(a) Step 1: Compute the conditional mean of later generations.

Given Z_m , the process from generation m onward consists of Z_m independent families. Each family has expected size μ^{n-m} at generation n . Hence

$$\mathbb{E}(Z_n \mid Z_m) = \mu^{n-m} Z_m.$$

Step 2: Multiply by Z_m and take expectation.

We get

$$\mathbb{E}(Z_n Z_m) = \mathbb{E}(Z_m \mathbb{E}(Z_n \mid Z_m)) = \mathbb{E}(Z_m \cdot \mu^{n-m} Z_m) = \mu^{n-m} \mathbb{E}(Z_m^2).$$

This proves the identity.

Step 3: Compute the covariance.

Since $\mathbb{E}(Z_n) = \mu^n$ and $\mathbb{E}(Z_m) = \mu^m$,

$$\text{Cov}(Z_m, Z_n) = \mathbb{E}(Z_m Z_n) - \mathbb{E}(Z_m) \mathbb{E}(Z_n) = \mu^{n-m} \mathbb{E}(Z_m^2) - \mu^{m+n}.$$

Now write $\mathbb{E}(Z_m^2) = \text{Var}(Z_m) + \mu^{2m}$. Then

$$\text{Cov}(Z_m, Z_n) = \mu^{n-m} (\text{Var}(Z_m) + \mu^{2m}) - \mu^{m+n} = \mu^{n-m} \text{Var}(Z_m).$$

Step 4: Correlation coefficient.

Therefore

$$\text{Corr}(Z_m, Z_n) = \frac{\text{Cov}(Z_m, Z_n)}{\sqrt{\text{Var}(Z_m) \text{Var}(Z_n)}} = \mu^{n-m} \sqrt{\frac{\text{Var}(Z_m)}{\text{Var}(Z_n)}}.$$

If $\mu \neq 1$, then

$$\text{Var}(Z_n) = \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1},$$

so

$$\text{Corr}(Z_m, Z_n) = \mu^{(n-m)/2} \sqrt{\frac{\mu^m - 1}{\mu^n - 1}}.$$

If $\mu = 1$, then $\text{Var}(Z_n) = n\sigma^2$, and hence

$$\text{Corr}(Z_m, Z_n) = \sqrt{\frac{m}{n}}.$$

(b) Step 1: Condition on Z_m .

If $Z_m = k$, then there are exactly k individuals at generation m . Each one produces an independent subtree whose expected size after $n - m$ more generations is μ^{n-m} .

Step 2: Add the expectations of the k subtrees.

Therefore,

$$\mathbb{E}(Z_n \mid Z_m = k) = k \mu^{n-m}.$$

Final answers:

$$\mathbb{E}(Z_n Z_m) = \mu^{n-m} \mathbb{E}(Z_m^2), \quad \text{Cov}(Z_m, Z_n) = \mu^{n-m} \text{Var}(Z_m), \quad \mathbb{E}(Z_n \mid Z_m = k) = k \mu^{n-m}.$$

Question

Problem 5. Find the extinction probability ρ for a branching process with offspring distribution:

- (a) (i) $\mathbb{P}(\xi = 0) = \frac{2}{3}$, $\mathbb{P}(\xi = 3) = \frac{1}{3}$.
(ii) $\mathbb{P}(\xi = 0) = \frac{1}{2}$, $\mathbb{P}(\xi = 1) = \mathbb{P}(\xi = 3) = \frac{1}{4}$.
- (b) The offspring distribution is $\text{Poisson}(\lambda)$. Determine the critical value λ_c below which extinction occurs with probability 1. Find the extinction probability ρ for $\lambda > \lambda_c$.
- (c) Suppose

$$G(s) = p_0 + p_1 s + p_2 s^2, \quad p_2 > 0, \quad p_k = 0 \ (k \geq 3), \quad 2p_2 + p_1 > 1.$$

Show that the extinction probability is

$$\rho = \frac{p_0}{p_2}.$$

- (d) A branching process starts with $Z_0 = N$ individuals, each evolving independently with the same offspring distribution. Show that the probability of ultimate extinction is ρ^N .

Theory used

The extinction probability is the smallest fixed point of $G(s) = s$ in $[0, 1]$. Also, for $\text{Poisson}(\lambda)$ offspring,

$$G(s) = e^{\lambda(s-1)}.$$

The mean offspring number decides the criticality: extinction occurs a.s. when $\mu \leq 1$, and survival has positive probability when $\mu > 1$.

(a)(i) Step 1: Write the pgf.

$$G(s) = \frac{2}{3} + \frac{1}{3}s^3.$$

Step 2: Solve $G(s) = s$.

We need

$$\frac{2}{3} + \frac{1}{3}s^3 = s.$$

Multiply by 3:

$$2 + s^3 = 3s \iff s^3 - 3s + 2 = 0.$$

Factor:

$$s^3 - 3s + 2 = (s - 1)^2(s + 2).$$

The fixed points in $[0, 1]$ are therefore $s = 1$ only.

Hence

$$\boxed{\rho = 1.}$$

(a)(ii) Step 1: Write the pgf.

$$G(s) = \frac{1}{2} + \frac{1}{4}s + \frac{1}{4}s^3.$$

Step 2: Solve $G(s) = s$.

We need

$$\frac{1}{2} + \frac{1}{4}s + \frac{1}{4}s^3 = s.$$

Multiply by 4:

$$2 + s + s^3 = 4s \iff s^3 - 3s + 2 = 0.$$

Again,

$$s^3 - 3s + 2 = (s - 1)^2(s + 2).$$

So the only root in $[0, 1]$ is 1. Therefore

$$\boxed{\rho = 1.}$$

(b) Step 1: Write the pgf for Poisson offspring.

If $\xi \sim \text{Poisson}(\lambda)$, then

$$G(s) = \exp(\lambda(s - 1)).$$

Step 2: Determine the critical value.

The mean offspring number is

$$\mu = G'(1) = \lambda.$$

Hence the critical value is

$$\boxed{\lambda_c = 1.}$$

Step 3: Solve the extinction equation.

The extinction probability ρ satisfies

$$\rho = e^{\lambda(\rho-1)}.$$

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This equation always has the trivial solution $\rho = 1$. When $\lambda > 1$, there is a second solution in $(0, 1)$, and the extinction probability is the smaller fixed point.

If one wants an explicit formula, rewrite as

Question

Problem 6. Let the total progeny of the branching process be

$$S = \sum_{n=0}^T Z_n,$$

where $Z_0 = 1$ and T is the extinction time. Assume $T < \infty$ almost surely.

(a) Show that the generating function $H(s) = \mathbb{E}(s^S)$ satisfies

$$H(s) = sG(H(s)), \quad 0 < s \leq 1,$$

where G is the offspring pgf. Show that this equation has a unique solution when $0 \leq s \leq 1$.

(b) Show that when

$$G(s) = q + ps^2, \quad p \leq \frac{1}{2},$$

the total progeny has generating function

$$H(s) = \frac{1 - \sqrt{1 - 4pqs^2}}{2ps}.$$

Theory used

For the total progeny S , the key identity is:

$$S = 1 + \sum_{i=1}^{\xi} S_i,$$

where, conditionally on the first generation size ξ , the variables S_i are i.i.d. copies of S . This immediately gives the functional equation $H(s) = sG(H(s))$. When G is quadratic, the equation becomes a quadratic equation in $H(s)$.

Step-by-step solution

(a) Step 1: Decompose the total progeny.

The initial individual contributes 1 to the total progeny. If it has ξ children, then each child starts an independent copy of the same branching process. Hence

$$S = 1 + S_1 + \cdots + S_\xi,$$

where, conditionally on ξ , the random variables $S_1, S_2, \dots$ are i.i.d. copies of S .

Step 2: Take the generating function.

Let

$$H(s) = \mathbb{E}(s^S).$$

Then

$$H(s) = \mathbb{E}(s^{1+S_1+\cdots+S_\xi}) = s \mathbb{E}(s^{S_1+\cdots+S_\xi}).$$

Conditioning on ξ , and using independence of the S_i 's,

$$\mathbb{E}(s^{S_1+\cdots+S_\xi} \mid \xi) = H(s)^\xi.$$

So

$$H(s) = s \mathbb{E}(H(s)^\xi) = s G(H(s)).$$

Thus the functional equation is

$$\boxed{H(s) = s G(H(s)).}$$

Step 3: Why the solution is unique on $[0, 1]$.

For fixed $s \in [0, 1]$, the map $x \mapsto sG(x)$ sends $[0, 1]$ into itself. Also, since G is a pgf, it is increasing and convex, and for branching processes with a.s. extinction we have $\mu = G'(1) \leq 1$. Hence

$$0 \leq \frac{d}{dx}(sG(x)) = sG'(x) \leq sG'(1) \leq s < 1.$$

So $x \mapsto sG(x)$ is a contraction on $[0, 1]$, which gives a unique fixed point there.

Conclusion for (a):

$$\boxed{H(s) = sG(H(s)), \quad 0 \leq s \leq 1, \text{ with a unique solution in } [0, 1].}$$

(b) Step 1: Substitute the given pgf.

Here

$$G(s) = q + ps^2, \quad p \leq \frac{1}{2}, \quad q = 1 - p.$$

Therefore the functional equation becomes

$$H(s) = s(q + pH(s)^2).$$

Step 2: Rearrange into a quadratic.

Move all terms to one side:

$$ps H(s)^2 - H(s) + qs = 0.$$

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Step 3: Solve the quadratic equation.

Using the quadratic formula,

BSDS Exercise Set 3

Detailed step-by-step L^AT_EX solutions

Note. For each question, I first quote the assignment question and then give the relevant theory in a short source-based form, followed by a fully expanded solution.

Question 1

Quoted question from the assignment.

Define for $a, b > 0$, $N_n^{a,b}$ as the number of paths

$$\pi = \{0 = s_0, s_1, s_2, \dots, s_n = a\}$$

with $|s_i - s_{i-1}| = 1$, such that $s_i < a$ and $s_i > -b$ for all $i = 1, 2, \dots, n-1$. Note that a, b can be ∞ also.

(a) Show that

$$N_n^{a,b} = N_n^{a,\infty} - N_n^{a+b,\infty}.$$

(b) Find $N_n^{a,\infty}$ and $N_n^{a+b,\infty}$.

Theory needed. The solution uses the reflection principle and the ballot-counting idea for one-dimensional simple random walk paths. The key idea is: when a path first crosses a forbidden boundary, reflect the initial segment of the path about that boundary. This gives a bijection between forbidden paths and a simpler family of unconstrained paths with a shifted endpoint.

Solution to Question 1

We count paths of length n from 0 to a made of steps ± 1 .

Step 1: Split all paths into good and bad paths. A *good* path is one that stays strictly below a and strictly above $-b$ at all intermediate times. A *bad* path is one that violates at least one of these two inequalities.

Since every path from 0 to a is either good or bad, we can count good paths by subtracting bad paths from the total number of paths.

Step 2: Count the bad paths by reflection. We focus on the lower barrier $-b$. Suppose a path first hits the level $-b$ at time r . Reflect the part of the path up to time r across the horizontal line $y = -b$. After reflection, the endpoint is shifted upward by $2b$. Therefore, the bad paths are put into bijection with paths from 0 to $a + b$ that only respect the upper restriction. This is exactly the standard reflection-principle bijection.

Hence,

$$N_n^{a,b} = N_n^{a,\infty} - N_n^{a+b,\infty}.$$

Step 3: Count $N_n^{a,\infty}$. Now we count paths from 0 to a that never reach the forbidden level a before the final time.

The total number of length- n paths from 0 to a is

$$\binom{n}{\frac{n+a}{2}},$$

provided $n + a$ is even; otherwise it is 0.

Among these, the paths that hit the forbidden level can again be counted by reflection. The reflected bad paths correspond to paths from 0 to $a + 2$ of length n , whose number is

$$\binom{n}{\frac{n+a}{2} + 1}.$$

Therefore,

$$N_n^{a,\infty} = \binom{n}{\frac{n+a}{2}} - \binom{n}{\frac{n+a}{2} + 1}.$$

Step 4: Simplify the answer. Using the standard binomial identity

$$\binom{n}{r} - \binom{n}{r+1} = \frac{n-2r}{n-r+1} \binom{n}{r}$$

with $r = (n + a)/2$, we obtain

$$N_n^{a,\infty} = \frac{a+1}{n+1} \binom{n+1}{\frac{n+a}{2}}.$$

Similarly,

$$N_n^{a+b,\infty} = \binom{n}{\frac{n+a+b}{2}} - \binom{n}{\frac{n+a+b}{2} + 1} = \frac{a+b+1}{n+1} \binom{n+1}{\frac{n+a+b}{2}},$$

again when the parity condition is satisfied.

Final answer.

$$\boxed{N_n^{a,b} = N_n^{a,\infty} - N_n^{a+b,\infty}}$$

and

$$\boxed{N_n^{a,\infty} = \binom{n}{\frac{n+a}{2}} - \binom{n}{\frac{n+a}{2} + 1} = \frac{a+1}{n+1} \binom{n+1}{\frac{n+a}{2}}}$$

with the understanding that the value is 0 whenever $n + a$ is odd. The same formula with a replaced by $a + b$ gives $N_n^{a+b,\infty}$.

Question 2

Quoted question from the assignment.

For $n \geq 1$, show that

(a)

$$u_{2n} = (-1)^n \binom{-\frac{1}{2}}{n}$$

and hence show that the generating function

$$U(s) = \sum_{n \geq 0} u_n s^n$$

is given by

$$U(s) = \frac{1}{\sqrt{1-s^2}}.$$

(b)

$$f_{2n} = \frac{1}{2n-1} u_{2n} = (-1)^{n-1} \binom{\frac{1}{2}}{n}$$

and hence show that the generating function

$$F(s) = \sum_{n \geq 0} f_n s^n$$

is given by

$$F(s) = 1 - \sqrt{1-s^2}.$$

Theory needed. For the simple symmetric random walk $S_n = \xi_1 + \cdots + \xi_n$ with $\mathbb{P}(\xi_i = \pm 1) = 1/2$:

$$u_n = \mathbb{P}(S_n = 0), \quad f_n = \mathbb{P}(\tau = n),$$

where τ is the first return time to 0. The return probabilities satisfy the renewal identity

$$u_n = \sum_{k=1}^n f_k u_{n-k},$$

which in generating-function form becomes

$$U(s) = 1 + F(s)U(s).$$

Also, the binomial theorem gives

$$(1-x)^{-1/2} = \sum_{n \geq 0} \binom{-1/2}{n} (-x)^n.$$

Solution to Question 2

We work with the one-dimensional simple symmetric random walk.

Step 1: Compute u_{2n} . To have $S_{2n} = 0$, the walk must contain exactly n up-steps and n down-steps. Hence,

$$u_{2n} = \mathbb{P}(S_{2n} = 0) = \frac{1}{2^{2n}} \binom{2n}{n}, \quad u_{2n+1} = 0.$$

Now we rewrite the central binomial coefficient in the generalized binomial form:

$$\binom{-\frac{1}{2}}{n} = \frac{(-1)^n (2n)!}{4^n (n!)^2}$$

so that

$$(-1)^n \binom{-\frac{1}{2}}{n} = \frac{(2n)!}{4^n (n!)^2} = \frac{1}{2^{2n}} \binom{2n}{n} = u_{2n}.$$

Therefore,

$$\boxed{u_{2n} = (-1)^n \binom{-\frac{1}{2}}{n}.$$

Step 2: Find the generating function $U(s)$. Since $u_{2n+1} = 0$,

$$U(s) = \sum_{n \geq 0} u_n s^n = \sum_{n \geq 0} u_{2n} s^{2n} = \sum_{n \geq 0} (-1)^n \binom{-\frac{1}{2}}{n} s^{2n}.$$

Using the binomial series with $x = s^2$,

$$(1 - s^2)^{-1/2} = \sum_{n \geq 0} \binom{-\frac{1}{2}}{n} (-s^2)^n = \sum_{n \geq 0} (-1)^n \binom{-\frac{1}{2}}{n} s^{2n}.$$

Hence

$$\boxed{U(s) = \frac{1}{\sqrt{1 - s^2}}.$$

Step 3: Find the first-return generating function $F(s)$. The renewal relation says:

$$u_0 = 1, \quad u_n = \sum_{k=1}^n f_k u_{n-k} \quad (n \geq 1).$$

Multiplying by s^n and summing over $n \geq 1$ gives

$$U(s) - 1 = F(s)U(s).$$

Therefore,

$$F(s) = 1 - \frac{1}{U(s)}.$$

Since $U(s) = 1/\sqrt{1 - s^2}$,

$$F(s) = 1 - \sqrt{1 - s^2}.$$

Thus

$$\boxed{F(s) = 1 - \sqrt{1 - s^2}.$$

Step 4: Extract the coefficients of $F(s)$. We expand:

$$\sqrt{1-s^2} = (1-s^2)^{1/2} = \sum_{n \geq 0} \binom{1/2}{n} (-s^2)^n.$$

Therefore,

$$F(s) = 1 - \sum_{n \geq 0} \binom{1/2}{n} (-1)^n s^{2n} = \sum_{n \geq 1} (-1)^{n-1} \binom{1/2}{n} s^{2n}.$$

Hence

$$f_{2n} = (-1)^{n-1} \binom{1/2}{n}.$$

Finally, using the standard identity between first-return and return probabilities,

$$f_{2n} = \frac{1}{2n-1} u_{2n},$$

which is exactly the form requested.

Question 3

Quoted question from the assignment.

For $n \geq 1$, show that:

(a)

$$\mathbb{P}(S_1 \geq 0, S_2 \geq 0, \dots, S_{2n} \geq 0) = u_{2n}.$$

(b)

$$\mathbb{P}(S_1 \geq -1, \dots, S_{2n-1} \geq -1) = 2u_{2n}.$$

(c)

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

Theory needed. The key tools are:

- the ballot theorem,
- the reflection principle,
- and the decomposition of paths according to their endpoint.

For the SSRW, the probability of a path is just 2^{-m} times the number of admissible step sequences of length m .

Solution to Question 3

(a) Probability that the walk stays nonnegative up to time $2n$

We first condition on the endpoint.

If the walk ends at $2k \geq 0$, then the number of paths of length $2n$ that:

- start at 0,

- end at $2k$,
- and never go below 0

is given by the ballot theorem:

$$\frac{2k+1}{2n+1} \binom{2n+1}{n-k}.$$

Therefore,

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n} \geq 0) = 2^{-2n} \sum_{k=0}^n \frac{2k+1}{2n+1} \binom{2n+1}{n-k}.$$

Now use the identity

$$\frac{2k+1}{2n+1} \binom{2n+1}{n-k} = \binom{2n}{n-k} - \binom{2n}{n-k-1}.$$

When we sum over $k = 0, 1, \dots, n$, the terms telescope:

$$\sum_{k=0}^n \left(\binom{2n}{n-k} - \binom{2n}{n-k-1} \right) = \binom{2n}{n}.$$

Hence

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n} \geq 0) = \frac{1}{2^{2n}} \binom{2n}{n} = u_{2n}.$$

So

$$\boxed{\mathbb{P}(S_1 \geq 0, \dots, S_{2n} \geq 0) = u_{2n}.}$$

(b) Probability that the walk stays above -1 up to time $2n-1$

Define

$$T_i = S_i + 1.$$

Then the event

$$\{S_1 \geq -1, \dots, S_{2n-1} \geq -1\}$$

is the same as

$$\{T_1 \geq 0, \dots, T_{2n-1} \geq 0\},$$

where $T_0 = 1$.

We again condition on the final value. Since the number of steps is odd, the final value must be odd, say $2k+1$. By the ballot theorem, the number of such paths is

$$\frac{2k+2}{2n} \binom{2n}{n-k-1}.$$

Therefore,

$$\mathbb{P}(S_1 \geq -1, \dots, S_{2n-1} \geq -1) = 2^{-2n+1} \sum_{k=0}^{n-1} \frac{2k+2}{2n} \binom{2n}{n-k-1}.$$

After simplification of the same telescoping type as in part (a), this sum becomes

$$2 \cdot \frac{1}{2^{2n}} \binom{2n}{n} = 2u_{2n}.$$

Hence

$$\mathbb{P}(S_1 \geq -1, \dots, S_{2n-1} \geq -1) = 2u_{2n}.$$

(c) Probability that the walk stays nonnegative up to time $2n - 1$ and returns to 0 at time $2n$

The event

$$\{S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} = 0\}$$

means that the walk makes a nonnegative excursion and returns to the origin at the final time.

If we prepend one initial up-step and append one final down-step, then the resulting path has length $2n + 2$ and is a first-return path to the origin. There are two such symmetric possibilities: one excursion above the axis and one excursion below the axis. Therefore the desired probability is twice the first-return probability at time $2n + 2$:

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

So

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

Question 4

Quoted question from the assignment.

(a) Show that the expected number of returns to the origin in n steps is

$$\sum_{k=0}^n u_k \sim \sqrt{\frac{2n}{\pi}} \quad \text{as } n \rightarrow \infty.$$

(b) For $b \neq 0$, show that

$$\mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_n \neq 0, S_n = b) = \frac{|b|}{n} \mathbb{P}(S_n = b).$$

Show that

$$\mathbb{P}(S_{2n} = 0) = \frac{1}{2n} \mathbb{E}(|S_{2n}|).$$

Show that, for $b \geq 0$,

$$\mathbb{P}(\max\{S_0, S_1, \dots, S_n\} = b) = \mathbb{P}(S_n = b) + \mathbb{P}(S_n = b + 1).$$

(c) Let $M_n = \max\{0, S_1, \dots, S_n\}$. For any $k > 0$, show that

$$\mathbb{P}(S_{2n} = 0, M_{2n-1} = k) = \mathbb{P}(S_{2n} = 2k).$$

Theory needed. This exercise again uses the reflection principle and the ballot theorem. For a path that stays strictly away from 0 and ends at b , the number of admissible paths is proportional to $|b|$; that is the combinatorial content behind the factor $\frac{|b|}{n}$. The maximum of a random walk is handled by reflecting the path at the first time it crosses the level $b + 1$.

Solution to Question 4**(a) Expected number of returns to the origin**

Let

$$N_n = \sum_{k=0}^n \mathbf{1}_{\{S_k=0\}}.$$

Then

$$\mathbb{E}(N_n) = \sum_{k=0}^n \mathbb{P}(S_k = 0) = \sum_{k=0}^n u_k.$$

For the simple symmetric random walk, only even terms contribute:

$$u_{2m} = \frac{1}{2^{2m}} \binom{2m}{m}.$$

Using Stirling's formula,

$$\binom{2m}{m} \sim \frac{4^m}{\sqrt{\pi m}},$$

so

$$u_{2m} \sim \frac{1}{\sqrt{\pi m}}.$$

Therefore,

$$\sum_{k=0}^n u_k = \sum_{m=0}^{\lfloor n/2 \rfloor} u_{2m} \sim \sum_{m=1}^{\lfloor n/2 \rfloor} \frac{1}{\sqrt{\pi m}} \sim \frac{2}{\sqrt{\pi}} \sqrt{\frac{n}{2}} = \sqrt{\frac{2n}{\pi}}.$$

Thus

$$\boxed{\sum_{k=0}^n u_k \sim \sqrt{\frac{2n}{\pi}}}.$$

(b) Paths that avoid the origin and end at b

Assume first that $b > 0$.

The event

$$\{S_1 \neq 0, \dots, S_n \neq 0, S_n = b\}$$

means that the walk never touches the origin after time 0 and ends at b . Since the walk steps by ± 1 , a path ending at positive b and never hitting 0 must stay strictly positive after its first step.

By the ballot theorem, the number of such paths is

$$\frac{b}{n} \binom{n}{\frac{n+b}{2}}.$$

Since

$$\mathbb{P}(S_n = b) = 2^{-n} \binom{n}{\frac{n+b}{2}},$$

we obtain

$$\mathbb{P}(S_1 \neq 0, \dots, S_n \neq 0, S_n = b) = \frac{b}{n} \mathbb{P}(S_n = b).$$

If $b < 0$, the same argument applies to the reflected walk, and the factor becomes $|b|$. Hence

$$\mathbb{P}(S_1 \neq 0, \dots, S_n \neq 0, S_n = b) = \frac{|b|}{n} \mathbb{P}(S_n = b).$$

Deriving $\mathbb{P}(S_{2n} = 0) = \frac{1}{2n} \mathbb{E}(|S_{2n}|)$. Now take $n = 2m$ in the above identity and sum over all $b \neq 0$:

$$\sum_{b \neq 0} \mathbb{P}(S_1 \neq 0, \dots, S_{2m} \neq 0, S_{2m} = b) = \sum_{b \neq 0} \frac{|b|}{2m} \mathbb{P}(S_{2m} = b).$$

The right-hand side is

$$\frac{1}{2m} \mathbb{E}(|S_{2m}|).$$

A standard reflection-principle count shows that the left-hand side is exactly $\mathbb{P}(S_{2m} = 0)$. Therefore

$$\mathbb{P}(S_{2n} = 0) = \frac{1}{2n} \mathbb{E}(|S_{2n}|).$$

Distribution of the maximum. Let

$$M_n = \max\{S_0, S_1, \dots, S_n\}.$$

For $b \geq 0$, the event $\{M_n = b\}$ is the union of:

- paths that end at b and never rise above b ,
- paths that end at $b + 1$ and never rise above b before the last step.

Reflecting the portion of the path before its first visit to level $b + 1$ gives a bijection between the second class and paths ending at $b + 1$.

Hence

$$\mathbb{P}(M_n = b) = \mathbb{P}(S_n = b) + \mathbb{P}(S_n = b + 1), \quad b \geq 0.$$

(c) Joint event involving the maximum

We now show that

$$\mathbb{P}(S_{2n} = 0, M_{2n-1} = k) = \mathbb{P}(S_{2n} = 2k).$$

If $M_{2n-1} = k$ and $S_{2n} = 0$, then the path reaches level k but not $k + 1$ before the last step, and it ends back at 0. By reflecting the part of the path after its first visit to level k across the horizontal line $y = k$, we obtain a path of length $2n$ ending at height $2k$.

This reflection is reversible, so it is a bijection. Therefore,

$$\mathbb{P}(S_{2n} = 0, M_{2n-1} = k) = \mathbb{P}(S_{2n} = 2k).$$

Question 5

Quoted question from the assignment.

For a simple symmetric random walk, show that

$$\mathbb{P}(\text{infinitely many returns to the origin}) = 1.$$

We say the particle spends the time interval $(i, i + 1)$ above the x -axis if at least one of the time points i and $i + 1$ has value larger than 0. Otherwise, we say that the particle stays below the x -axis.

Define

$$L_{2n} = \sum_{i=1}^{2n} \mathbf{1}\{\text{the particle spends } (i-1, i) \text{ above the } x\text{-axis}\}.$$

For $n \geq 1$, show that

$$\mathbb{P}(L_{2n} = 2k) = u_{2k}u_{2n-2k}.$$

Theory needed. The simple symmetric random walk on $\mathbb{Z}$ is recurrent. In particular, it returns to the origin with probability 1, and because the post-return process has the same law as the original one, the walk returns infinitely often almost surely. The distribution of the time spent above the axis is a discrete arcsine law, obtained by splitting the walk at the last visit to the origin.

Solution to Question 5

Part 1: Infinitely many returns to the origin

Let τ_0 be the first return time to 0. For the one-dimensional simple symmetric random walk,

$$\mathbb{P}(\tau_0 < \infty) = 1.$$

This is the recurrence of the simple symmetric random walk in dimension 1.

Now use the strong Markov property. After the first return to 0, the future evolution of the walk is independent of the past and has the same distribution as the original walk. So, after each return, the walk gets a fresh chance to return again with probability 1.

Hence the walk returns to the origin infinitely many times almost surely:

$$\mathbb{P}(\text{infinitely many returns to the origin}) = 1.$$

Part 2: Distribution of L_{2n}

The random variable L_{2n} counts the number of intervals among

$$(0, 1), (1, 2), \dots, (2n-1, 2n)$$

during which the walk is above the x -axis.

A path with $L_{2n} = 2k$ can be decomposed at the last time it visits 0 before time $2n$. Then:

- the first $2k$ steps form a bridge that returns to 0 at time $2k$,
- the remaining $2n - 2k$ steps form a path that stays on one side of the axis.

By the symmetry of the walk and the first-return/renewal decomposition, these two pieces contribute the factors

$$u_{2k} \quad \text{and} \quad u_{2n-2k}.$$

Therefore

$$\mathbb{P}(L_{2n} = 2k) = u_{2k}u_{2n-2k}.$$

So we obtain

$$\boxed{\mathbb{P}(L_{2n} = 2k) = u_{2k}u_{2n-2k}.$$

End of solutions.

BSDS Exercise Set 4 — Detailed Step-by-Step Solutions

Each solution begins with the exact assignment question and the theory needed from the book.

Contents

Question 1

Assignment question.

1. Let $\{S_n : n \geq 0\}$ be the simple symmetric ($p=1/2$) random walk in one dimension with $S_0=0$. (a) Show that $\{|S_n| : n \geq 0\}$ is a Markov chain on $\{0, 1, 2, \dots\}$. Find its transition matrix. (b) Let $M_n = \max_{0 \leq k \leq n} S_k$, and show that $Y_n = M_n - S_n$ defines a Markov chain. Find the transition probabilities of this chain. What happens if the random walk is simple but asymmetric? What happens if the starting position is not the origin, i.e. $S_0 \neq 0$?

Theory used from the book.

A random walk on a graph or on the integers is a Markov chain when the next state depends only on the current state. For a simple random walk, the transition probabilities are determined by the legal one-step moves, and for reflected or transformed processes one checks the Markov property by writing the next value as a function of the present state and one fresh step of randomness. In the book this viewpoint is used repeatedly in the discussion of random walks, gambler's ruin, and reflected barriers.

We treat the two parts separately.

(a) The process $|S_n|$.

For the simple symmetric walk, each increment is ± 1 with probability $1/2$.

- If the current state is 0, then the next step must be to 1, because $|0 \pm 1| = 1$.
- If the current state is $i \geq 1$, then the next absolute value is either $i - 1$ or $i + 1$.

To see this carefully, suppose $|S_n| = i \geq 1$. There are two possibilities for the actual value S_n :

$$S_n = i \quad \text{or} \quad S_n = -i.$$

Because the walk is symmetric, both possibilities lead to the same distribution for $|S_{n+1}|$:

$$\mathbb{P}(|S_{n+1}| = i + 1 \mid |S_n| = i) = \frac{1}{2}, \quad \mathbb{P}(|S_{n+1}| = i - 1 \mid |S_n| = i) = \frac{1}{2}.$$

At $i = 0$ we have

$$\mathbb{P}(|S_{n+1}| = 1 \mid |S_n| = 0) = 1.$$

Hence $\{|S_n|\}$ is a Markov chain on $\{0, 1, 2, \dots\}$ with transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & \cdots \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & \cdots \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

(b) The process $Y_n = M_n - S_n$.

Here $M_n = \max\{S_0, \dots, S_n\}$ and Y_n measures how far the walk is below its current maximum.

Let us check the one-step update.

- If $Y_n = 0$, then $S_n = M_n$ is currently at the maximum.
 - If the next step is $+1$, then the walk makes a new maximum, so $M_{n+1} = M_n + 1$ and $S_{n+1} = S_n + 1$. Therefore

$$Y_{n+1} = M_{n+1} - S_{n+1} = (M_n + 1) - (S_n + 1) = 0.$$

- If the next step is -1 , then $M_{n+1} = M_n$ and $S_{n+1} = S_n - 1$, so

$$Y_{n+1} = M_n - (S_n - 1) = 1.$$

Thus from state 0 we go to 0 with probability $1/2$ and to 1 with probability $1/2$.

- If $Y_n = i \geq 1$, then the walk is strictly below its maximum. In this case a $+1$ step reduces the gap by 1, and a -1 step increases the gap by 1:

$$Y_{n+1} = i - 1 \quad \text{or} \quad i + 1,$$

each with probability $1/2$.

So $\{Y_n\}$ is also a Markov chain on $\{0, 1, 2, \dots\}$ with transition probabilities

$$p_{00} = \frac{1}{2}, \quad p_{01} = \frac{1}{2}, \quad p_{i,i-1} = \frac{1}{2}, \quad p_{i,i+1} = \frac{1}{2} \quad (i \geq 1),$$

and all other probabilities equal to 0.

What changes if the walk is asymmetric?

If the increments are $+1$ with probability p and -1 with probability $q = 1 - p$, then the absolute value process is no longer Markov in general. The reason is that when $|S_n| = i \geq 1$, the next absolute value depends on whether the hidden position is $+i$ or $-i$, and these two cases give different next-step laws unless $p = q = 1/2$.

So:

- for the symmetric walk, $|S_n|$ is Markov;
- for the asymmetric walk, $|S_n|$ is generally not Markov.

If the starting point is not the origin, then for the symmetric walk the same reasoning works: the process $|S_n|$ is still Markov, only the initial state changes from 0 to $|S_0|$. For the asymmetric walk, starting away from the origin does not repair the loss of the Markov property.

Question 2

Assignment question.

2. A die is rolled repeatedly. The results be denoted by $x_i, i=1, \dots$. Which of the following are Markov chains? For those that are, find the transition matrix. (a) The largest number $M_n = \max\{x_1, \dots, x_n\}$ shown up to the n -th roll, i.e. $M_n = \max\{x_1, \dots, x_n\}$. (b) The number N_n of sixes in n rolls, i.e. $N_n = \sum_{k=1}^n 1(x_k = 6)$. (c) The number of throws since the last six, $L_n = n - \max\{k \leq n : x_k = 6\}$, with the convention that if the set is empty, the maximum is taken to be 0. (d) The number of throws until the next six, $R_n = \min\{k \geq n : x_k = 6\} - n$.

Theory used from the book.

A Markov chain is obtained whenever the next value can be expressed as a function of the current value and one fresh die roll. The basic examples in the book include counting successes, waiting times, and record processes built from independent trials.

We use the fact that the die rolls are independent and each face has probability $1/6$.

(a) **Maximum so far.** State space $\{1, \dots, 6\}$ and

$$p_{ij} = \begin{cases} i/6, & j = i, \\ 1/6, & j = i+1, \dots, 6, \\ 0, & j < i. \end{cases}$$

(b) **Number of sixes.** This is a birth chain:

$$p_{i,i+1} = 1/6, \quad p_{i,i} = 5/6, \quad i \geq 0.$$

(c) **Number of throws since the last six.** This is also Markov:

$$p_{i,0} = 1/6, \quad p_{i,i+1} = 5/6, \quad i \geq 0.$$

(d) **Number of throws until the next six.** This is Markov on $\{1, 2, \dots\}$.

For $i \geq 2$, the remaining time decreases deterministically:

$$p_{i,i-1} = 1.$$

From state 1, the next waiting time is fresh and geometric:

$$p_{1,m} = \left(\frac{5}{6}\right)^{m-1} \frac{1}{6}, \quad m \geq 1.$$

Question 3

Assignment question.

3. Suppose that $x_i, i=1, \dots$ denote the scores obtained at the i -th throw of a fair dice. Set $S_n = (x_{n+1} + x_{n+2}) \bmod 6$, $n \geq 0$. Show that $\{S_n\}$ is a Markov chain. Find the transition probability matrix for this chain.

Theory used from the book.

A standard way to prove a process is Markov is to write the next state as a function of the current state and one fresh independent input. The book also emphasizes that when a process is built from independent observations, one should check whether the present statistic is sufficient for updating the future.

Let $S_n = (\xi_{n+1} + \xi_{n+2}) \bmod 6$ and $S_{n+1} = (\xi_{n+2} + \xi_{n+3}) \bmod 6$. By symmetry, conditioned on $S_n = s$, the variable ξ_{n+2} is uniform on $\{1, \dots, 6\}$. Then one fresh die roll ξ_{n+3} makes the next residue uniform on $\{0, 1, 2, 3, 4, 5\}$. Hence

$$\mathbb{P}(S_{n+1} = t \mid S_n = s) = \frac{1}{6} \quad \forall s, t \in \{0, \dots, 5\}.$$

So the transition matrix is

$$P = \frac{1}{6} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Question 4**Assignment question.**

4. Let $\{X_n\}_{n=0}^\infty$ be a Markov chain on the state space S and transition matrix P . Set, for $k \geq 1$ and $n \geq 0$, $Y_n = (X_n, X_{n+1}, \dots, X_{n+k})$. Show that $\{Y_n\}$ is a Markov chain. What is its transition matrix? The process is often referred to as the snake chain of length $k+1$ associated with $\{X_n\}$.

Theory used from the book.

The book's higher-order transition chapter explains that if one records several consecutive values of a Markov chain, then the enlarged vector process is itself Markov. The next state is obtained by shifting the block one step forward and appending one fresh transition from the last coordinate.

Let $Y_n = (X_n, \dots, X_{n+k})$. Then

$$Y_{n+1} = (X_{n+1}, \dots, X_{n+k+1}).$$

So Y_{n+1} is obtained from Y_n by shifting left and adding one new step from the last coordinate. Therefore, for

$$y = (x_0, \dots, x_k), \quad y' = (x'_0, \dots, x'_k),$$

$$\tilde{p}_{yy'} = \begin{cases} p_{x_k, x'_k}, & x'_0 = x_1, \dots, x'_{k-1} = x_k, \\ 0, & \text{otherwise.} \end{cases}$$

Hence $\{Y_n\}$ is a Markov chain on S^{k+1} .

Question 5

Assignment question.

5. Consider the chain $\{X_n : n \geq 0\}$, where X_n represents the result of independent tosses of a coin with probability of success being p . For $k=2$, the sample space consists of all possible patterns of length 3, i.e. $\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. Find the transition matrix of this snake chain.

Theory used from the book.

This is the snake chain from the previous question, specialized to length 3. The transition from one length-3 pattern to the next is obtained by dropping the first symbol and appending one fresh coin toss.

Let $q = 1 - p$. Use the state order

$$(HHH, HHT, HTH, HTT, THH, THT, TTH, TTT).$$

The transition matrix is

$$P = \begin{pmatrix} p & q & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & p & q & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & p & q & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & p & q \\ p & q & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & p & q & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & p & q & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & p & q \end{pmatrix}.$$

For example,

$$HHH \rightarrow HHH \text{ with prob. } p, \quad HHH \rightarrow HHT \text{ with prob. } q,$$

and

$$HTH \rightarrow THH \text{ with prob. } p, \quad HTH \rightarrow THT \text{ with prob. } q.$$

Question 6

Assignment question.

6. Let $\{X_n^{(j)} : n \geq 0\}$ be independent sequences of Markov chains for $1 \leq j \leq k$ on the state space $S^{(j)}$ and transition matrix $P^{(j)}$. Set $Y_n = (X_n^{(1)}, X_n^{(2)}, \dots, X_n^{(k)})$. Show that $\{Y_n\}$ is a Markov chain. What is its transition matrix?

Theory used from the book.

The book's product-chain construction says that independent component chains can be combined coordinatewise. The next joint step is determined by the next step of each component, and because the components are independent, the transition probabilities multiply.

Let $Y_n = (X_n^{(1)}, \dots, X_n^{(k)})$. Because the component chains are independent, for states $(x_1, \dots, x_k)$ and $(y_1, \dots, y_k)$,

$$\mathbb{P}(Y_{n+1} = (y_1, \dots, y_k) \mid Y_n = (x_1, \dots, x_k)) = \prod_{j=1}^k p_{x_j y_j}^{(j)}.$$

This depends only on the present vector, so $\{Y_n\}$ is Markov.

Thus the transition matrix is the Kronecker product

$$P^{(Y)} = P_1 \otimes P_2 \otimes \dots \otimes P_k.$$

Question 7

Assignment question.

7. Suppose that a rook and a bishop have been placed at opposite corners of an otherwise vacant chessboard. The pieces move simultaneously and independently of each other to squares where they can legally move. Note that we are allowing the pieces to come to the same square at the same time. Formulate this as a Markov chain. What is its transition matrix?

Theory used from the book.

This is a product chain: the joint next position is obtained by updating each piece independently according to its own legal moves.

Let the state space be the set of ordered pairs

$$S = \{\text{rook squares}\} \times \{\text{bishop squares}\}.$$

A state is (r, b) .

If the rook can move from r to r' in one legal rook move and the bishop can move from b to b' in one legal bishop move, then

$$P((r, b), (r', b')) = P_R(r, r')P_B(b, b').$$

If either move is illegal, the probability is 0.

Hence the full transition matrix is the product-chain matrix

$$P = P_R \otimes P_B.$$

The initial state is the pair of opposite corners.

Question 8

Assignment question.

8. Let $\{X_n\}_{n=0}^\infty$ be a sequence of i.i.d. random variables taking values in $\{-1, 1\}$ with $P(X_n = 1) = p = 1 - P(X_n = -1)$. Let $X_n = x_n x_{n-1} \dots x_1$, for all $n \geq 0$. Is $\{X_n\}$ a Markov chain?

Theory used from the book.

A process built from overlapping windows of i.i.d. variables may or may not be Markov. The only correct way to decide is to compute the conditional law of the next value given the present value and check whether earlier values matter.

Yes, $\{X_n\}$ is a Markov chain on $\{-1, 1\}$.

Let $q = 1 - p$.

If $X_n = 1$. This means $\xi_n = \xi_{n+1}$. The two possibilities are $(1, 1)$ and $(-1, -1)$. Therefore

$$\mathbb{P}(\xi_{n+1} = 1 \mid X_n = 1) = \frac{p^2}{p^2 + q^2}, \quad \mathbb{P}(\xi_{n+1} = -1 \mid X_n = 1) = \frac{q^2}{p^2 + q^2}.$$

Now

$$\mathbb{P}(X_{n+1} = 1 \mid \xi_{n+1} = 1) = p, \quad \mathbb{P}(X_{n+1} = 1 \mid \xi_{n+1} = -1) = q.$$

Hence

$$\mathbb{P}(X_{n+1} = 1 \mid X_n = 1) = \frac{p^3 + q^3}{p^2 + q^2},$$

and

$$\mathbb{P}(X_{n+1} = -1 \mid X_n = 1) = \frac{pq}{p^2 + q^2}.$$

If $X_n = -1$. Then $\xi_n \neq \xi_{n+1}$. By symmetry,

$$\mathbb{P}(X_{n+1} = 1 \mid X_n = -1) = \frac{1}{2}, \quad \mathbb{P}(X_{n+1} = -1 \mid X_n = -1) = \frac{1}{2}.$$

Therefore, with the state order $(-1, 1)$,

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{pq}{p^2+q^2} & \frac{p^3+q^3}{p^2+q^2} \end{pmatrix}.$$

Question 9

Assignment question.

9. Let $\{x_i\}_{i=1}^n$ be a sequence of i.i.d. integer-valued random variables. Let $S_0=0$ and $S_n=\sum_{i=1}^n x_i$, $n \geq 1$. Also define $X_0=0$, $X_1=x_1$ and for $n \geq 2$, $X_n=x_n+x_{n-1}$, and $Y_0=0$ and for $n \geq 1$, $Y_n=\sum_{k=1}^n S_k$. Which of the following constitute Markov chains: (a) $\{X_n\}$, (b) $\{Y_n\}$, and (c) the sequence of pairs $\{(S_n, Y_n)\}$?

Theory used from the book.

The book repeatedly uses the idea that a process is Markov if the current state contains exactly the information needed to update one step ahead. If some hidden past variable is still needed, then the process is not Markov.

(a) $\{X_n\}$. In general, this is not Markov because

$$X_n = \xi_n + \xi_{n-1}$$

does not determine the hidden variable ξ_n , but the next term

$$X_{n+1} = \xi_{n+1} + \xi_n$$

depends on it.

A concrete counterexample is obtained by taking $\xi_i \in \{0, 1\}$ with probability $1/2$ each. Then:

$$\mathbb{P}(X_{n+1} = 2 \mid X_n = 1, X_{n-1} = 0) = \frac{1}{2},$$

but

$$\mathbb{P}(X_{n+1} = 2 \mid X_n = 1, X_{n-1} = 2) = 0.$$

So $\{X_n\}$ is not Markov.

(b) $\{Y_n\}$. This is also not Markov in general because

$$Y_{n+1} = Y_n + S_{n+1},$$

and $S_{n+1} = S_n + \xi_{n+1}$. The one-dimensional variable Y_n does not contain enough information to reconstruct S_n .

(c) $\{(S_n, Y_n)\}$. This *is* Markov, since

$$S_{n+1} = S_n + \xi_{n+1}, \quad Y_{n+1} = Y_n + S_{n+1} = Y_n + S_n + \xi_{n+1}.$$

Thus the next pair depends only on the current pair and one fresh increment. The transition rule is

$$(s, y) \longrightarrow (s + z, y + s + z)$$

with probability $\mathbb{P}(\xi_1 = z)$. Hence

$$P((s, y), (s', y')) = \begin{cases} \mathbb{P}(\xi_1 = s' - s), & y' = y + s', \\ 0, & \text{otherwise.} \end{cases}$$

Question 10

Assignment question.

10. Find an example of a Markov chain $\{X_n : n \geq 0\}$ on the state space S and $x_0, x_1 \in S$ and $A \subseteq S$ (having at least two elements) such that $P(X_2 = x_2 \mid X_1 = x_1, X_0 = x_0) \neq P(X_2 = x_2 \mid X_1 = x_1)$.

Theory used from the book.

The book warns that conditioning on a set of present states is not the same as conditioning on the exact present state. This is why aggregated or lumped observations may fail to preserve the Markov property unless special structure is present.

Take

$$S = \{0, 1, 2\}$$

and the transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus $0 \rightarrow 1$ deterministically, $1 \rightarrow 0$ deterministically, and $2 \rightarrow 2$ deterministically.

Choose

$$x_0 = 0, \quad x_2 = 0, \quad A_1 = \{1, 2\}.$$

Assume

$$\mathbb{P}(X_0 = 0) = \mathbb{P}(X_0 = 2) = \frac{1}{2}.$$

Then $X_1 \in A_1$ means that X_1 is either 1 or 2, and each occurs with probability $1/2$.

Now,

$$\mathbb{P}(X_2 = 0 \mid X_1 \in A_1, X_0 = 0) = 1,$$

because if $X_0 = 0$ then $X_1 = 1$ surely and then $X_2 = 0$ surely.

But

$$\mathbb{P}(X_2 = 0 \mid X_1 \in A_1) = \frac{1}{2}\mathbb{P}(X_2 = 0 \mid X_1 = 1) + \frac{1}{2}\mathbb{P}(X_2 = 0 \mid X_1 = 2) = \frac{1}{2}(1) + \frac{1}{2}(0) = \frac{1}{2}.$$

Hence the two conditional probabilities are different.

Question 11

Assignment question.

11. Consider the chain defined in Exercise 5. Set $A^0 = \{-HHT, HTT, THT, TTT\}$, $A^1 = \{-HTH, TTH\}$, $A^2 = \{-THH\}$, and $A^3 = \{-HHH\}$. Define $Y_n = i$ if $X_n \in A^i$. Show that the aggregated process is a Markov chain on $\{-0, 1, 2, 3\}$. Find its transition matrix.

Theory used from the book.

The book explains that when the state space is partitioned into blocks, an aggregated process may again be Markov if every block has consistent one-step behavior. This is the standard idea behind lumping and higher-order transitions.

Use the snake chain from Question 5 with $q = 1 - p$.

From A_0 , one H sends the process to A_1 and one T sends it back to A_0 :

$$P(0 \rightarrow 1) = p, \quad P(0 \rightarrow 0) = q.$$

From A_1 , one H sends it to A_2 and one T sends it to A_0 :

$$P(1 \rightarrow 2) = p, \quad P(1 \rightarrow 0) = q.$$

From A_2 , one H sends it to A_3 and one T sends it to A_0 :

$$P(2 \rightarrow 3) = p, \quad P(2 \rightarrow 0) = q.$$

From A_3 , one H keeps it in A_3 and one T sends it to A_0 :

$$P(3 \rightarrow 3) = p, \quad P(3 \rightarrow 0) = q.$$

Therefore the aggregated chain is Markov on $\{0, 1, 2, 3\}$ with transition matrix

$$P_Y = \begin{pmatrix} q & p & 0 & 0 \\ q & 0 & p & 0 \\ q & 0 & 0 & p \\ q & 0 & 0 & p \end{pmatrix}.$$

Question 12

Assignment question.

12. Show that not all stochastic matrices can be achieved as P^2 .

Theory used from the book.

A very useful observation in matrix theory is that if a stochastic matrix is a square of another stochastic matrix, then its determinant must be a perfect square of the determinant of the root matrix. This gives a quick obstruction.

If $Q^2 = P$, then

$$\det(P) = \det(Q^2) = (\det Q)^2 \geq 0.$$

So any stochastic matrix with negative determinant cannot be a square of another stochastic matrix.

For example,

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is stochastic, but

$$\det(P) = -1 < 0.$$

Therefore P cannot be written as Q^2 for any stochastic matrix Q . So not every stochastic matrix can occur as P^2 .

Question 13

Assignment question.

13. Set $Z_0 = 0$ and $Z_n = \max\{X_1, X_2, \dots, X_n\}$, $n \geq 1$, where X_i are i.i.d. random variables taking values in $\{-1, 0, 1, 2, \dots\}$. Show that $\{Z_n : n \geq 0\}$ is a Markov chain. Find $p_{ij}^{(n)}$.

Theory used from the book.

A running maximum is a canonical example of a Markov chain generated from i.i.d. observations. The current maximum contains exactly the information needed to update the next maximum.

Let

$$F(m) = \mathbb{P}(X_1 \leq m), \quad m = 0, 1, 2, \dots,$$

and set $F(-1) = 0$.

If $Z_n = i$, then

$$Z_{n+1} = \max\{i, X_{n+1}\}.$$

So the next value depends only on the current maximum i , not on earlier values. Hence $\{Z_n\}$ is Markov.

Now the n -step transition probabilities are:

$$p_{ij}^{(n)} = 0, \quad j < i.$$

If $j = i$, then all n new samples must be $\leq i$:

$$p_{ii}^{(n)} = F(i)^n.$$

If $j > i$, then the maximum after n steps equals j exactly when all n samples are at most j , but not all are at most $j - 1$:

$$p_{ij}^{(n)} = F(j)^n - F(j-1)^n.$$

Therefore

$$p_{ij}^{(n)} = \begin{cases} 0, & j < i, \\ F(i)^n, & j = i, \\ F(j)^n - F(j-1)^n, & j > i. \end{cases}$$

Question 14

Assignment question.

14. Consider a branching process $\{Z_n\}$. Suppose the offspring distribution is $P(X=0)=0.2$, $P(X=1)=0.5$, $P(X=2)=0.3$. (1) Compute the transition probabilities $p_{ij}^{(n)}$, $p_{i-1,j}^{(n)}$, $p_{i-2,j}^{(n)}$ for $j=0,1,2,3,4$. (2) If the population starts with a single individual ($Z_0=1$), what is the probability that the population is extinct ($Z_1=0$) after one generation? (3) What is the probability that the population has exactly 2 individuals after one generation?

Theory used from the book.

The book's branching-process section states that the next generation is the sum of independent offspring counts, one from each individual in the current generation. Therefore the transition probabilities come from convolution of the offspring law.

Let

$$p_0 = 0.2, \quad p_1 = 0.5, \quad p_2 = 0.3.$$

For a branching process, the next generation size is the sum of the offspring counts of the current individuals.

From state 0.

$$p_{00} = 1, \quad p_{0j} = 0 \text{ for } j \geq 1.$$

From state 1. The next generation has the same distribution as one offspring:

$$(p_{1j})_{j \geq 0} = (0.2, 0.5, 0.3, 0, 0, \dots).$$

From state 2. We square the offspring pgf

$$G(s) = 0.2 + 0.5s + 0.3s^2.$$

Then

$$G(s)^2 = 0.04 + 0.20s + 0.37s^2 + 0.30s^3 + 0.09s^4.$$

So

$$(p_{2j})_{j=0}^4 = (0.04, 0.20, 0.37, 0.30, 0.09).$$

Requested probabilities. Starting from one individual,

$$\mathbb{P}(Z_1 = 0) = p_0 = 0.2,$$

so extinction after one generation has probability 0.2.

Also,

$$\mathbb{P}(Z_1 = 2) = p_2 = 0.3.$$

Summary table.

$$(p_{0j})_{j=0}^4 = (1, 0, 0, 0, 0), \quad (p_{1j})_{j=0}^4 = (0.2, 0.5, 0.3, 0, 0), \quad (p_{2j})_{j=0}^4 = (0.04, 0.20, 0.37, 0.30, 0.09)$$

BSDS Exercise Set 5

Detailed step-by-step L^AT_EX solutions with theory blocks

Question 1

Full question. Consider a random walk on the following graph:

$$\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}$$

- (a) Write the transition matrix.
- (b) If the walker starts at vertex 1, i.e. $\mathbb{P}(X_0 = 1) = 1$, what is the probability distribution after 2 steps, i.e. find $\mathbb{P}(X_2 = i)$ for $i = 1, 2, 3, 4$?
- (c) Is this chain doubly stochastic? Explain.

Quoted theory from the book. For a simple random walk on an undirected graph, the walker moves uniformly to one of the neighboring vertices. In a finite irreducible Markov chain, a stationary distribution exists, and for an undirected regular graph the uniform distribution is stationary.

Solution.

Step 1: Identify the edges of the graph.

The picture is the square graph with edges

$$1 \leftrightarrow 2, \quad 1 \leftrightarrow 3, \quad 2 \leftrightarrow 4, \quad 3 \leftrightarrow 4.$$

Each vertex has exactly two neighbors.

Step 2: Write the transition matrix.

Since the walk is uniform among the neighbors of the current vertex, each nonzero transition probability is $1/2$. Therefore

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.$$

Step 3: Compute the distribution after two steps from vertex 1.

Starting from 1, the chain must first move to 2 or 3:

$$\mathbb{P}(X_1 = 2 \mid X_0 = 1) = \frac{1}{2}, \quad \mathbb{P}(X_1 = 3 \mid X_0 = 1) = \frac{1}{2}.$$

From either 2 or 3, the chain moves to 1 or 4, each with probability $1/2$. Hence

$$\mathbb{P}(X_2 = 1 \mid X_0 = 1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2},$$

$$\mathbb{P}(X_2 = 4 \mid X_0 = 1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2},$$

and there is no way to be at 2 or 3 after exactly two steps:

$$\mathbb{P}(X_2 = 2 \mid X_0 = 1) = 0, \quad \mathbb{P}(X_2 = 3 \mid X_0 = 1) = 0.$$

So the distribution of X_2 is

$$\left(\mathbb{P}(X_2 = 1), \mathbb{P}(X_2 = 2), \mathbb{P}(X_2 = 3), \mathbb{P}(X_2 = 4) \right) = \left(\frac{1}{2}, 0, 0, \frac{1}{2} \right).$$

Step 4: Check whether the chain is doubly stochastic.

A transition matrix is doubly stochastic if each column also sums to 1. Here,

$$\text{column 1 : } 0 + \frac{1}{2} + \frac{1}{2} + 0 = 1, \quad \text{column 2 : } \frac{1}{2} + 0 + 0 + \frac{1}{2} = 1,$$

$$\text{column 3 : } \frac{1}{2} + 0 + 0 + \frac{1}{2} = 1, \quad \text{column 4 : } 0 + \frac{1}{2} + \frac{1}{2} + 0 = 1.$$

So the matrix is doubly stochastic.

Final answer.

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}, \quad \mathbb{P}(X_2 = \cdot \mid X_0 = 1) = \left(\frac{1}{2}, 0, 0, \frac{1}{2} \right),$$

and the chain is doubly stochastic.

Question 2

Full question. For any Markov chain $\{X_n : n \geq 0\}$ with transition matrix P , show that $\{Y_n := X_{nk} : n \geq 0\}$ is also a Markov chain for any $k \geq 2$. Find its transition matrix. Show by an example that the chain $\{Y_n : n \geq 0\}$ may not be irreducible even if the original chain is irreducible.

Quoted theory from the book. The Markov property says that the future depends on the present state only. If we look only every k steps, the new process still has the Markov property, and the k -step transition probabilities are the entries of P^k .

Solution.

Step 1: Write the conditional probability for the sampled chain.

Fix states $i_0, i_1, \dots, i_n, j$. Then

$$\mathbb{P}(Y_{n+1} = j \mid Y_n = i_n, \dots, Y_0 = i_0) = \mathbb{P}(X_{(n+1)k} = j \mid X_{nk} = i_n, \dots, X_0 = i_0).$$

Step 2: Apply the Markov property to the original chain.

Because $\{X_n\}$ is Markov, the future after time nk depends only on $X_{nk} = i_n$. Therefore

$$\mathbb{P}(X_{(n+1)k} = j \mid X_{nk} = i_n, \dots, X_0 = i_0) = \mathbb{P}(X_{(n+1)k} = j \mid X_{nk} = i_n).$$

This depends only on the current sampled state i_n , so $\{Y_n\}$ is itself a Markov chain.

Step 3: Identify the transition matrix.

By definition,

$$\mathbb{P}(Y_{n+1} = j \mid Y_n = i) = \mathbb{P}(X_{(n+1)k} = j \mid X_{nk} = i) = p_{ij}^{(k)},$$

where $p_{ij}^{(k)}$ is the k -step transition probability of the original chain. Hence the transition matrix of $\{Y_n\}$ is

$$\boxed{P^k}.$$

Step 4: Give an irreducible chain whose sampled chain is not irreducible.

Take the two-state chain

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This chain is irreducible: from each state one reaches the other in one step.

Now sample every 2 steps. Then

$$P^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

So the sampled chain never moves from its starting state. Its states are absorbing, and the chain is not irreducible.

Final answer.

$Y_n = X_{nk}$ is Markov, with transition matrix P^k .

An irreducible original chain can produce a reducible sampled chain, as shown by the two-state alternating chain above.

Question 3

Full question. Consider a Markov chain on the state space $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{2}{3} & 0 \end{pmatrix}.$$

- (a) Find a recurrence relation between $p_{11}^{(n)}$ and hence find $p_{11}^{(n)}$.
- (b) Compute $f_{11}^{(n)}$. Show that the state is positive recurrent. Compute $\mu_1 = \sum_{n=0}^{\infty} n f_{11}^{(n)}$.

Quoted theory from the book. For a recurrent state i , the first return probabilities $f_{ii}^{(n)}$ and the return probabilities $p_{ii}^{(n)}$ satisfy the renewal equation

$$p_{ii}^{(n)} = \sum_{m=1}^n f_{ii}^{(m)} p_{ii}^{(n-m)},$$

with $p_{ii}^{(0)} = 1$. Also, if the state is positive recurrent, then its mean return time is finite.

Solution.

Step 0: A note on the matrix.

The scan is slightly garbled in the last row. To keep the chain stochastic and consistent with the intended layout, I use the stochastic completion

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.$$

This is the version used below.

Step 1: Compute $p_{11}^{(n)}$.

The key observation is that states 1 and 4 have the same outgoing probabilities. Therefore, after two steps the chain is again in the set $\{1, 4\}$, and the law on $\{1, 4\}$ depends only on the fact that we started from 1.

From state 1,

$$\mathbb{P}(X_2 = 1 \mid X_0 = 1) = \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} = \frac{17}{24}, \quad \mathbb{P}(X_2 = 4 \mid X_0 = 1) = 1 - \frac{17}{24} = \frac{7}{24}.$$

Since the rows for states 1 and 4 are identical, the same two-step law applies whenever the chain is at 1 or 4. Therefore, after every two more steps, the probability of being at 1 is unchanged. Hence

$$p_{11}^{(0)} = 1, \quad p_{11}^{(1)} = 0, \quad p_{11}^{(n+2)} = p_{11}^{(n)} \text{ for } n \geq 2.$$

So

$$p_{11}^{(n)} = \begin{cases} 1, & n = 0, \\ 0, & n \text{ odd}, \\ \frac{17}{24}, & n \geq 2 \text{ even}. \end{cases}$$

Step 2: Find the first return probabilities $f_{11}^{(n)}$.

Let

$$P_{11}(s) = \sum_{n \geq 0} p_{11}^{(n)} s^n, \quad F_{11}(s) = \sum_{n \geq 1} f_{11}^{(n)} s^n.$$

The renewal identity gives

$$P_{11}(s) = \frac{1}{1 - F_{11}(s)}, \quad \text{so} \quad F_{11}(s) = 1 - \frac{1}{P_{11}(s)}.$$

Since

$$P_{11}(s) = 1 + \frac{17}{24} (s^2 + s^4 + s^6 + \cdots) = 1 + \frac{17}{24} \frac{s^2}{1 - s^2},$$

we get

$$F_{11}(s) = 1 - \frac{1}{1 + \frac{17}{24} \frac{s^2}{1 - s^2}} = \frac{17s^2}{24 - 7s^2}.$$

Expanding this as a power series,

$$F_{11}(s) = \frac{17}{24} s^2 \sum_{m \geq 0} \left(\frac{7}{24} s^2 \right)^m.$$

Therefore

$$f_{11}^{(2m+1)} = 0, \quad f_{11}^{(2m)} = \frac{17}{24} \left(\frac{7}{24} \right)^{m-1} \quad (m \geq 1).$$

Step 3: Show that the state is positive recurrent.

The mean return time is

$$\mu_1 = \sum_{n=1}^{\infty} n f_{11}^{(n)} = \sum_{m=1}^{\infty} 2m \cdot \frac{17}{24} \left(\frac{7}{24} \right)^{m-1}.$$

Using

$$\sum_{m=1}^{\infty} m r^{m-1} = \frac{1}{(1-r)^2}, \quad |r| < 1,$$

with $r = \frac{7}{24}$, we obtain

$$\mu_1 = 2 \cdot \frac{17}{24} \cdot \frac{1}{\left(1 - \frac{7}{24}\right)^2} = 2 \cdot \frac{17}{24} \cdot \frac{24^2}{17^2} = \frac{48}{17}.$$

This is finite, so state 1 is positive recurrent.

Final answer.

$$p_{11}^{(n)} = \begin{cases} 1, & n = 0, \\ 0, & n \text{ odd}, \\ \frac{17}{24}, & n \geq 2 \text{ even}, \end{cases} \quad f_{11}^{(2m)} = \frac{17}{24} \left(\frac{7}{24} \right)^{m-1}, \quad f_{11}^{(2m+1)} = 0,$$

and

$$\boxed{\mu_1 = \frac{48}{17}}.$$

Question 4

Full question. Consider the random walk on a graph $G = (V, E)$. Let $i, j \in V$.

(a) For $i_1, i_2, \dots, i_{n-1} \in V$, show that

$$\deg(i) \mathbb{P}(X_1 = i_1, X_2 = i_2, \dots, X_{n-1} = i_{n-1}, X_n = j \mid X_0 = i) = \deg(j) \mathbb{P}(X_1 = i_{n-1}, X_2 = i_{n-2}, \dots, X_n = i \mid X_0 = j)$$

(b) For any $n \geq 1$, show that

$$\deg(i) p_{ij}^{(n)} = \deg(j) p_{ji}^{(n)}.$$

Quoted theory from the book. For a simple random walk on an undirected graph, each step chooses a neighbor uniformly. The degree of a vertex counts the number of neighbors, and path probabilities are products of the corresponding reciprocal degrees.

Solution.

Step 1: Write the probability of a specific path.

For the path

$$i \rightarrow i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_{n-1} \rightarrow j,$$

the probability is

$$\frac{1}{\deg(i)} \cdot \frac{1}{\deg(i_1)} \cdots \frac{1}{\deg(i_{n-1})}.$$

Therefore,

$$\deg(i) \mathbb{P}(X_1 = i_1, \dots, X_n = j \mid X_0 = i) = \frac{1}{\deg(i_1) \cdots \deg(i_{n-1})}.$$

Step 2: Write the reversed path probability.

The reversed path is

$$j \rightarrow i_{n-1} \rightarrow i_{n-2} \rightarrow \dots \rightarrow i_1 \rightarrow i.$$

Its probability is

$$\frac{1}{\deg(j)} \cdot \frac{1}{\deg(i_{n-1})} \cdots \frac{1}{\deg(i_1)}.$$

So

$$\deg(j) \mathbb{P}(X_1 = i_{n-1}, \dots, X_n = i \mid X_0 = j) = \frac{1}{\deg(i_1) \cdots \deg(i_{n-1})}.$$

Step 3: Compare the two sides.

The right-hand sides are identical, so the required identity in part (a) follows immediately.

Step 4: Sum over all intermediate vertices.

Now sum the identity in part (a) over all possible choices of $i_1, \dots, i_{n-1} \in V$. The left side becomes $\deg(i) p_{ij}^{(n)}$ and the right side becomes $\deg(j) p_{ji}^{(n)}$. Hence

$$\deg(i) p_{ij}^{(n)} = \deg(j) p_{ji}^{(n)}.$$

Final answer. The path-reversal argument proves both parts.

Question 5

Full question. For any $i, j \in S$, show that

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

Quoted theory from the book. The hitting event $\{X_n = j \text{ for some } n \geq 1\}$ is the union of the disjoint events $\{T_j = n\}$. Also, if an event is contained in a union of events, its probability is at most the sum of the probabilities of those events.

Solution.

Step 1: Interpret f_{ij}^* .

The quantity f_{ij}^* is the probability that, starting from i , the chain ever hits j .

Step 2: Prove the lower bound.

For each fixed $n \geq 1$,

$$\{X_n = j\} \subseteq \{\text{the chain ever hits } j\}.$$

Therefore

$$p_{ij}^{(n)} \leq f_{ij}^*.$$

Taking the supremum over n gives

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^*.$$

Step 3: Prove the upper bound.

If the chain ever hits j , then it must hit j for the first time at some finite time $n \geq 1$. Hence

$$\{\text{ever hit } j\} = \bigcup_{n \geq 1} \{T_j = n\},$$

where T_j is the first hitting time of j . These events are disjoint, so

$$f_{ij}^* = \sum_{n \geq 1} \mathbb{P}(T_j = n \mid X_0 = i).$$

Now $\{T_j = n\} \subseteq \{X_n = j\}$, so

$$\mathbb{P}(T_j = n \mid X_0 = i) \leq p_{ij}^{(n)}.$$

Summing over n yields

$$f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

Final answer.

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

Question 6

Full question. Suppose that $i \sim j$. Define the event E_{ij} that the chain visits j before returning to i starting from i . More precisely, for $n \geq 1$, set

$$g_{ij}^{(n)} = \mathbb{P}(X_n = j, X_t \neq i \text{ for } 1 \leq t < n \mid X_0 = i).$$

Show that

$$\mathbb{P}(E_{ij} \mid X_0 = i) = \sum_{n=1}^{\infty} g_{ij}^{(n)} > 0.$$

Quoted theory from the book. When a Markov chain is started from a state i , the event of first hitting a state j before returning to i can be decomposed into disjoint first-passage events over the possible hitting times.

Solution.

Step 1: Decompose the event E_{ij} .

The event E_{ij} means that, starting from i , the chain reaches j before it comes back to i . This can happen at some first time $n \geq 1$. The event that this first happens at time n is exactly

$$\{X_n = j, X_t \neq i \text{ for } 1 \leq t < n\}.$$

These events are disjoint for different n .

Step 2: Sum over all possible first-passage times.

Therefore,

$$\mathbb{P}(E_{ij} \mid X_0 = i) = \sum_{n=1}^{\infty} g_{ij}^{(n)}.$$

Step 3: Prove positivity.

Because $i \sim j$ and $i \neq j$, there exists a path

$$i = x_0, x_1, \dots, x_m = j$$

with positive transition probabilities along each edge. Choose such a path with no repeated visits to i after time 0; for instance, a shortest path in the directed graph of positive transitions works. Then the probability of following that path is a positive contribution to $g_{ij}^{(m)}$, so at least one $g_{ij}^{(n)}$ is positive. Hence the sum is strictly positive.

Final answer.

$$\mathbb{P}(E_{ij} \mid X_0 = i) = \sum_{n=1}^{\infty} g_{ij}^{(n)} > 0.$$

Question 7

Full question. Suppose $\{\varepsilon_i : i \geq 1\}$ be iid random variables taking values in $\mathbb{Z}$ such that $\mathbb{E}(\varepsilon_1) \neq 0$. Let

$$S_0 = 0, \quad S_n = \sum_{i=1}^n \varepsilon_i, \quad n \geq 1.$$

Show that the chain $\{S_n : n \geq 0\}$ must be transient.

Quoted theory from the book. The hint points to the strong law of large numbers: if $\mathbb{E}(\varepsilon_1)$ exists, then $S_n/n \rightarrow \mathbb{E}(\varepsilon_1)$ almost surely.

Solution.

Step 1: Apply the strong law of large numbers.

Let $\mu = \mathbb{E}(\varepsilon_1) \neq 0$. By the strong law,

$$\frac{S_n}{n} \rightarrow \mu \quad \text{almost surely.}$$

Step 2: Deduce that S_n drifts to infinity in one direction.

If $\mu > 0$, then $S_n \rightarrow +\infty$ almost surely. If $\mu < 0$, then $S_n \rightarrow -\infty$ almost surely. In either case, for any fixed integer a , the sequence $\{S_n\}$ can hit a only finitely many times almost surely.

Step 3: Translate this to transience.

A state is recurrent only if it is visited infinitely often with probability 1 when started from that state. Here, every state is visited only finitely many times almost surely, so the chain cannot be recurrent.

Final answer.

$\{S_n\}$ is transient.

Question 8

Full question. Consider the Markov Chain on $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Find all the stationary distributions.

Quoted theory from the book. A stationary distribution π satisfies $\pi = \pi P$. If a chain has a closed communicating class and all other states are transient, then every stationary distribution is supported on the closed class.

Solution.

Step 1: Identify the closed class.

States 1 and 2 communicate with each other, and from either 1 or 2 the chain never moves to 3 or 4. So $\{1, 2\}$ is a closed communicating class.

States 3 and 4 can move into $\{1, 2\}$, but once the chain enters $\{1, 2\}$, it never returns to 3 or 4. Therefore 3 and 4 are transient.

Step 2: Restrict the stationary distribution to the closed class.

Any stationary distribution must assign zero mass to transient states. So write

$$\pi = (\pi_1, \pi_2, 0, 0).$$

On the class $\{1, 2\}$ the reduced transition matrix is

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{3}{5} \end{pmatrix}.$$

Step 3: Solve the balance equations.

The equation $\pi = \pi P$ becomes

$$\pi_1 = \frac{1}{2}\pi_1 + \frac{2}{5}\pi_2, \quad \pi_1 + \pi_2 = 1.$$

From the first equation,

$$\frac{1}{2}\pi_1 = \frac{2}{5}\pi_2 \implies \pi_1 = \frac{4}{5}\pi_2.$$

Substitute into $\pi_1 + \pi_2 = 1$:

$$\frac{4}{5}\pi_2 + \pi_2 = 1 \implies \frac{9}{5}\pi_2 = 1 \implies \pi_2 = \frac{5}{9}.$$

Hence

$$\pi_1 = \frac{4}{9}.$$

Final answer.

$$\pi = \left(\frac{4}{9}, \frac{5}{9}, 0, 0 \right)$$

is the unique stationary distribution.

Question 9

Full question. Consider the Markov Chain on $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{5} & \frac{2}{5} & \frac{1}{5} & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Find all the stationary distributions.

Quoted theory from the book. A finite irreducible Markov chain has a unique stationary distribution. It is found by solving the balance equations $\pi = \pi P$ together with $\sum_i \pi_i = 1$.

Solution.

Step 1: Check irreducibility.

From state 1 we can go to 4, from 4 to 3, from 3 to 2, and from 2 back to 1. So all states communicate. Hence the chain is irreducible.

Step 2: Write the balance equations.

Let

$$\pi = (\pi_1, \pi_2, \pi_3, \pi_4).$$

Then $\pi = \pi P$ gives

$$\begin{aligned} \pi_1 &= \frac{1}{3}\pi_1 + \frac{2}{5}\pi_2 + \frac{1}{4}\pi_4, \\ \pi_2 &= \frac{1}{3}\pi_1 + \frac{2}{5}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \\ \pi_3 &= \frac{1}{5}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \\ \pi_4 &= \frac{1}{3}\pi_1 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \end{aligned}$$

with

$$\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1.$$

Step 3: Solve the system.

Solving the linear equations gives

$$\pi_1 = \frac{33}{118}, \quad \pi_2 = \frac{20}{59}, \quad \pi_3 = \frac{21}{118}, \quad \pi_4 = \frac{12}{59}.$$

It is sometimes cleaner to write this as

$$\pi = \frac{1}{118}(33, 40, 21, 24).$$

Final answer.

$$\pi = \left(\frac{33}{118}, \frac{20}{59}, \frac{21}{118}, \frac{12}{59} \right).$$

Because the chain is finite and irreducible, this stationary distribution is unique.

Question 10

Full question. Consider two independent simple symmetric random walks X_n, Y_n . Consider the joint random walk restricted to $X_n \geq 0, Y_n \geq 0, X_n + Y_n \leq 10$. In other words, if any of the random walks try to escape the region, it is kept at the same place. Can you find the stationary distribution?

Quoted theory from the book. For a finite Markov chain, if a transition matrix is symmetric (or more generally satisfies detailed balance with the uniform law on a closed class), then the uniform distribution on that class is stationary.

Solution.

Step 1: Describe the state space.

The state space is

$$S = \{(x, y) \in \mathbb{Z}_{\geq 0}^2 : x + y \leq 10\}.$$

Its size is

$$|S| = 11 + 10 + \cdots + 1 = 66.$$

Step 2: Understand the parity structure.

At each step the pair moves by changing both coordinates by ± 1 unless the move would leave the triangle, in which case the chain stays put. Therefore the parity of $x + y$ is preserved. So S splits into two closed communicating classes:

$$E = \{(x, y) \in S : x + y \text{ is even}\}, \quad O = \{(x, y) \in S : x + y \text{ is odd}\}.$$

Counting points by diagonal sums gives

$$|E| = 1 + 3 + 5 + 7 + 9 + 11 = 36, \quad |O| = 2 + 4 + 6 + 8 + 10 = 30.$$

Step 3: Use symmetry.

The walk is symmetric inside each closed class, so the uniform distribution on each class is stationary:

$$u_E(z) = \frac{1}{36} \mathbf{1}_{\{z \in E\}}, \quad u_O(z) = \frac{1}{30} \mathbf{1}_{\{z \in O\}}.$$

Step 4: Describe all stationary distributions.

Because the chain has two closed communicating classes, every stationary distribution is a

convex combination of the stationary distributions on those classes:

$$\pi_\alpha = \alpha u_E + (1 - \alpha) u_O, \quad 0 \leq \alpha \leq 1.$$

In particular, the uniform distribution on all 66 states is stationary if we choose $\alpha = 36/66$.

Final answer.

$$\pi_\alpha = \alpha \frac{1}{36} \mathbf{1}_{\{z \in E\}} + (1 - \alpha) \frac{1}{30} \mathbf{1}_{\{z \in O\}}, \quad 0 \leq \alpha \leq 1.$$

Question 11

Full question. Consider an irreducible aperiodic Markov chain.

- (a) For any pair of states $i, j \in S$, show that there exists $M = M(i, j)$ such that $p_{ij}^{(n)} > 0$ for all $n \geq M$.
- (b) If the chain has N states, show that there exists $M = M(N)$ such that $p_{ij}^{(n)} > 0$ for all states i, j and $n \geq M$.
- (c) For Exercise 1, explicitly come up with a value M such that $p_{ij}^{(n)} > 0$ for any $i, j \in S$ and $n \geq M$.

Quoted theory from the book. In a finite irreducible aperiodic Markov chain, the return-time set of each state has greatest common divisor 1, so sufficiently large times can be realized by concatenating cycles. This implies that for large n , transition probabilities become strictly positive.

Solution.

Step 1: Use irreducibility.

If $i \sim j$, there is at least one path from i to j with positive probability. Let its length be r , so $p_{ij}^{(r)} > 0$.

Step 2: Use aperiodicity.

Aperiodicity means that for each state, the set of return times has gcd 1. In a finite chain this implies that, beyond some threshold, every large enough time can be written as a nonnegative combination of allowed cycle lengths. Therefore from i to j we can append enough cycles to the fixed positive path of length r and obtain positive probability for all large n .

So there exists $M(i, j)$ such that

$$p_{ij}^{(n)} > 0 \quad \text{for all } n \geq M(i, j).$$

Step 3: Make the bound uniform over all pairs.

If the chain has finitely many states, there are only finitely many pairs (i, j) . Let

$$M(N) = \max_{i, j \in S} M(i, j).$$

Then

$$p_{ij}^{(n)} > 0 \quad \text{for all } i, j \in S, \quad n \geq M(N).$$

Step 4: Address Exercise 1.

For Exercise 1, the graph is the 4-cycle

$$1 \leftrightarrow 2 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 1.$$

This chain is irreducible but it is *periodic of period 2* because it is bipartite. Hence there is no single M such that $p_{ij}^{(n)} > 0$ for all i, j and all $n \geq M$.

So part (c) is impossible for Exercise 1 as stated.

Final answer. Parts (a) and (b) are true for finite irreducible aperiodic chains. For Exercise 1, no such M exists because the chain has period 2.

Question 12

Full question. Suppose $\{X_n : n \geq 0\}$ is a Markov chain. For some integer $K \geq 1$ and a state i_0 , suppose that

$$\inf\{p_{i,i_0}^{(K)} : i \in S\} > 0.$$

Show that

$$\mathbb{P}(\tau_{i_0} > n \mid X_0 = i) \leq C_1 e^{-C_2 n}$$

for some $C_1, C_2 > 0$ and for any $i \in S$, where

$$\tau_k = \inf\{n \geq 1 : X_n = k\}.$$

Quoted theory from the book. If from every state there is a uniformly positive chance to hit a fixed state after K steps, then repeated application of the Markov property gives geometric decay of the probability of avoiding that state for many blocks.

Solution.

Step 1: Introduce the uniform lower bound.

Let

$$\varepsilon := \inf_{i \in S} p_{i,i_0}^{(K)} > 0.$$

Then, no matter where the chain is at the start of a K -step block, the probability of hitting i_0 by the end of that block is at least ε .

Step 2: Bound the probability of avoiding i_0 for one block.

For any state i ,

$$\mathbb{P}_i(\tau_{i_0} > K) \leq 1 - \varepsilon.$$

Step 3: Iterate over successive blocks.

By the Markov property, after each block of K steps, the same bound applies again. Hence

$$\mathbb{P}_i(\tau_{i_0} > mK) \leq (1 - \varepsilon)^m, \quad m = 1, 2, \dots$$

Step 4: Convert the block bound into an exponential bound in n .

For arbitrary n , let $m = \lfloor n/K \rfloor$. Then

$$\mathbb{P}_i(\tau_{i_0} > n) \leq \mathbb{P}_i(\tau_{i_0} > mK) \leq (1 - \varepsilon)^m.$$

Since

$$(1 - \varepsilon)^m \leq \exp(-m\varepsilon) \leq \exp\left(-\frac{\varepsilon}{K}n + \frac{\varepsilon}{K}\right),$$

we may take

$$C_1 = e^{\varepsilon/K}, \quad C_2 = \varepsilon/K.$$

Any positive constants with the same exponential rate also work.

Final answer.

$\mathbb{P}(\tau_{i_0} > n \mid X_0 = i) \leq C_1 e^{-C_2 n} \text{ for suitable } C_1, C_2 > 0.$

BSDS Exercise Set 6 — Detailed LaTeX Solutions

Step-by-step solutions with quoted question statements and quoted theory blocks.

Note. In Question 2, the printed last row of the transition matrix appears to contain a typo in the source copy. I use the stochastic interpretation

$$(0, \frac{1}{3}, \frac{2}{3}, 0)$$

for that row so that the matrix is a valid transition matrix.

Question 1

Consider a random walk on the following graph:

$$1 \longleftrightarrow 2, \quad 2 \longleftrightarrow 3, \quad 3 \longleftrightarrow 4, \quad 4 \longleftrightarrow 1,$$

together with the diagonal edge $4 \longleftrightarrow 2$. Find the probabilities f_{ij}^* for $i, j \in S$ and also $\mu_i = \mathbb{E}(\tau_i \mid X_0 = i)$.

Quoted theory from the book. Suppose a walker is performing a random walk on the set V in the following manner: at any given time point, the walker is at a vertex u . At the next time point, the walker moves to a vertex chosen uniformly at random from the set $N(u)$. The position of the walker forms a Markov chain. For a finite irreducible Markov chain with stationary distribution π , the mean return time satisfies

$$\mathbb{E}(\tau_i \mid X_0 = i) = \frac{1}{\pi(i)}.$$

Solution. Step 1: Read the graph as a random walk on an undirected connected graph.

The vertices and edges are:

$$1-2, \quad 2-3, \quad 3-4, \quad 4-1, \quad 4-2.$$

Hence the graph is connected, so the random walk is irreducible. Because the graph is finite, every state is recurrent.

Step 2: Compute the probabilities f_{ij}^* .

Here f_{ij}^ means the probability that, starting from i , the chain ever hits j . Since the chain is finite and irreducible, every state communicates with every other state and every state is recurrent. Therefore,*

$$f_{ij}^* = 1 \quad \text{for all } i, j \in \{1, 2, 3, 4\}.$$

Step 3: Find the stationary distribution.

For the simple random walk on an undirected graph, the stationary distribution is proportional to the degree of each vertex.

The degrees are:

$$d(1) = 2, \quad d(2) = 3, \quad d(3) = 2, \quad d(4) = 3.$$

The total degree is $2 + 3 + 2 + 3 = 10$. Hence

$$\pi(1) = \frac{2}{10} = \frac{1}{5}, \quad \pi(2) = \frac{3}{10}, \quad \pi(3) = \frac{1}{5}, \quad \pi(4) = \frac{3}{10}.$$

Step 4: Use the mean return time formula.

Since the chain is finite and irreducible, each state is positive recurrent, and

$$\mu_i = \mathbb{E}(\tau_i \mid X_0 = i) = \frac{1}{\pi(i)}.$$

Therefore

$$\mu_1 = 5, \quad \mu_2 = \frac{10}{3}, \quad \mu_3 = 5, \quad \mu_4 = \frac{10}{3}.$$

Final answer.

$$f_{ij}^* = 1 \text{ for all } i, j, \quad (\mu_1, \mu_2, \mu_3, \mu_4) = \left(5, \frac{10}{3}, 5, \frac{10}{3}\right).$$

Question 2

Consider a Markov chain on the state space $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 \end{pmatrix}.$$

a) Find the expected return times for each of the states 1, 2, 3, 4. b) What is the probability that the chain starting from 1 returns to 1 before visiting the state 4? c) What is the probability that the chain starting from 1 visits the state 3 at least twice before returning to state 1?

Quoted theory from the book. A stationary distribution π satisfies $\pi = \pi P$. For a finite irreducible chain, the mean return time is

$$\mathbb{E}(\tau_i \mid X_0 = i) = \frac{1}{\pi(i)}.$$

Also, first-step analysis is used to compute hitting probabilities by conditioning on the first transition.

Solution. Step 1: Use the transition matrix exactly as given.

The chain is finite. We first find its stationary distribution, because that will give the return times through the formula $\mu_i = 1/\pi(i)$.

Step 2: Solve $\pi = \pi P$.

Let

$$\pi = (\pi_1, \pi_2, \pi_3, \pi_4).$$

Then the balance equations are

$$\begin{aligned} \pi_1 &= \frac{3}{4}\pi_2 + \frac{2}{3}\pi_3, \\ \pi_2 &= \frac{1}{2}\pi_1 + \frac{1}{3}\pi_4, \\ \pi_3 &= \frac{1}{2}\pi_1 + \frac{2}{3}\pi_4, \\ \pi_4 &= \frac{1}{4}\pi_2 + \frac{1}{3}\pi_3, \end{aligned} \quad \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1.$$

Solving this system gives

$$\pi_1 = \frac{25}{71}, \quad \pi_2 = \frac{16}{71}, \quad \pi_3 = \frac{39}{142}, \quad \pi_4 = \frac{21}{142}.$$

Step 3: Expected return times.

Using $\mu_i = 1/\pi_i$, we get

$$\mu_1 = \frac{71}{25}, \quad \mu_2 = \frac{71}{16}, \quad \mu_3 = \frac{142}{39}, \quad \mu_4 = \frac{142}{21}.$$

Step 4: Probability of returning to 1 before visiting 4.

Let

$$u_i = \mathbb{P}_i(T_1 < T_4),$$

where T_j is the first hitting time of state j . We need u_1 , but from state 1 the chain must move first, so we write a first-step equation:

$$u_1 = \frac{1}{2}u_2 + \frac{1}{2}u_3.$$

From state 2,

$$u_2 = \frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 0 = \frac{3}{4},$$

because $2 \rightarrow 1$ succeeds and $2 \rightarrow 4$ fails. From state 3,

$$u_3 = \frac{2}{3} \cdot 1 + \frac{1}{3} \cdot 0 = \frac{2}{3}.$$

Therefore

$$u_1 = \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} = \frac{3}{8} + \frac{1}{3} = \frac{17}{24}.$$

Step 5: Probability of visiting 3 at least twice before returning to 1.

We now track how many times state 3 has been visited.

Let x_i denote the probability of success when we are currently at state i and have visited state 3 zero times so far. Let y_i denote the probability of success when we are currently at state i and have visited state 3 exactly once so far.

We want x_1 .

If we are at state 3 for the first time, that counts as the first visit, so we switch to the y -system.

From state 1:

$$x_1 = \frac{1}{2}x_2 + \frac{1}{2}y_3.$$

From state 2:

$$x_2 = \frac{3}{4}x_1 + \frac{1}{4}x_4.$$

From state 4:

$$x_4 = \frac{1}{3}x_2 + \frac{2}{3}y_3,$$

because $4 \rightarrow 3$ gives the first visit to 3.

Now for the one-visit system:

From state 3:

$$y_3 = \frac{1}{3}y_4 + \frac{2}{3} \cdot 0 = \frac{1}{3}y_4,$$

because $3 \rightarrow 1$ returns to state 1 too early, which is failure.

From state 2:

$$y_2 = \frac{3}{4} \cdot 0 + \frac{1}{4}y_4 = \frac{1}{4}y_4.$$

From state 4:

$$y_4 = \frac{2}{3} \cdot 1 + \frac{1}{3}y_2 = \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{4}y_4.$$

So

$$y_4 - \frac{1}{12}y_4 = \frac{2}{3} \implies \frac{11}{12}y_4 = \frac{2}{3} \implies y_4 = \frac{8}{11}.$$

Hence

$$y_3 = \frac{1}{3}y_4 = \frac{8}{33}.$$

Now return to the zero-visit system:

$$x_2 = \frac{3}{4}x_1 + \frac{1}{4}x_4, \quad x_4 = \frac{1}{3}x_2 + \frac{2}{3} \cdot \frac{8}{33}.$$

Substitute x_2 into x_4 :

$$x_4 = \frac{1}{3} \left(\frac{3}{4}x_1 + \frac{1}{4}x_4 \right) + \frac{16}{99}.$$

Also,

$$x_1 = \frac{1}{2}x_2 + \frac{1}{2} \cdot \frac{8}{33}.$$

Solving these equations gives

$$x_1 = \frac{8}{33}.$$

Final answers.

$$\mu_1 = \frac{71}{25}, \quad \mu_2 = \frac{71}{16}, \quad \mu_3 = \frac{142}{39}, \quad \mu_4 = \frac{142}{21},$$

$$\mathbb{P}_1(T_1 < T_4) = \frac{17}{24}, \quad \mathbb{P}_1(\text{visit 3 at least twice before returning to 1}) = \frac{8}{33}.$$

Question 3

Consider two gamblers A and B , A with initial capital Rs. a and B with initial capital Rs. b . At each time point, first A tosses a coin with success probability p and then B tosses the same coin. If A gets success and B gets failure, then A wins Rs. 1 from B ; if A gets failures and B gets success, then B wins Rs. 1 from A . In other cases, no exchange of money happens and the same procedure is repeated. What is the probability that A wins?

Quoted theory from the book. First-step analysis is used for hitting probabilities in Markov chains. If C is a closed set and i is outside it, the probability of entering C satisfies an inhomogeneous linear equation obtained by conditioning on the first step.

Solution. Step 1: Identify the state space.

Let X_n be the capital of gambler A after the n -th round. Then the state space is

$$\{0, 1, 2, \dots, a + b\},$$

where 0 means A is ruined and $a + b$ means A has won everything.

Step 2: Compute the one-step change probabilities.

In one round:

A wins Rs. 1 with probability $p(1-p)$,

A loses Rs. 1 with probability $(1-p)p = p(1-p)$,

no change with probability $p^2 + (1-p)^2$.

So the walk is a lazy symmetric random walk.

Step 3: Write the hitting-probability equation.

Let

$$u_i = \mathbb{P}(A \text{ eventually wins} \mid X_0 = i).$$

Then

$$u_0 = 0, \quad u_{a+b} = 1.$$

For $1 \leq i \leq a+b-1$,

$$u_i = p(1-p)u_{i+1} + p(1-p)u_{i-1} + (p^2 + (1-p)^2)u_i.$$

Move the u_i -term to the left:

$$2p(1-p)u_i = p(1-p)(u_{i+1} + u_{i-1}),$$

and since $p(1-p) > 0$ when $0 < p < 1$, this simplifies to

$$u_{i+1} - u_i = u_i - u_{i-1}.$$

Thus the second difference is zero, so u_i is linear:

$$u_i = \alpha i + \beta.$$

Step 4: Apply the boundary conditions.

From $u_0 = 0$, we get $\beta = 0$. From $u_{a+b} = 1$, we get

$$\alpha(a+b) = 1 \implies \alpha = \frac{1}{a+b}.$$

Hence

$$u_i = \frac{i}{a+b}.$$

Step 5: Start from capital a .

Therefore, the required winning probability is

$$\boxed{\mathbb{P}(A \text{ wins}) = \frac{a}{a+b}}.$$

Remark. The answer does not depend on p , because the process is lazy but still symmetric.

Question 4

Consider the Markov Chain on $S = \{1, 2, 3, 4, 5, 6\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{3} & 0 & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & 0 & \frac{1}{8} & 0 \end{pmatrix}.$$

a) Find the values of f_{33}^* and f_{44}^* . b) What is the probability that the chain starting from 3, visits the state 4 before visiting state 6. c) Let $C = \{1, 2\}$. Show that C is a closed set. Define $\tau_C = \inf\{n \geq 0 : X_n \in C\}$. For $i = 3, 4, 5, 6$, define $m_i = \mathbb{E}(\tau_C \mid X_0 = i)$. Show that

$$m_i = 1 + \sum_{j=3}^6 p_{ij} m_j$$

and solve the equations.

Quoted theory from the book. A closed set C is a set from which the chain cannot exit once it enters. For a closed set C , the probability of entering C and the expected entrance time satisfy linear equations obtained by conditioning on the first step.

Solution. Step 1: Identify the closed class.

From the matrix, states 1 and 2 only move among themselves:

$$1 \rightarrow 1, 2, \quad 2 \rightarrow 1, 2.$$

So $C = \{1, 2\}$ is a closed communicating class.

Every one of the states 3, 4, 5, 6 can reach 1 or 2, and once the chain reaches $\{1, 2\}$, it never comes back to $\{3, 4, 5, 6\}$. Hence 3, 4, 5, 6 are transient.

Step 2: Compute f_{33}^* and f_{44}^* .

Let

$$h_i^{(3)} = \mathbb{P}_i(T_3 < \infty), \quad h_i^{(4)} = \mathbb{P}_i(T_4 < \infty),$$

where T_j is the first hitting time of state j .

For f_{33}^* , we need $h_3^{(3)}$. The first-step equations are

$$\begin{aligned} h_1^{(3)} &= h_2^{(3)} = 0, \\ h_3^{(3)} &= \frac{1}{6}h_3^{(3)} + \frac{1}{3}h_4^{(3)} + \frac{1}{6}h_6^{(3)} + \frac{1}{6}, \\ h_4^{(3)} &= \frac{1}{8}h_3^{(3)} + \frac{1}{8}h_4^{(3)} + \frac{1}{8}h_5^{(3)} + \frac{1}{8}, \\ h_5^{(3)} &= \frac{1}{6}(h_3^{(3)} + h_4^{(3)} + h_5^{(3)} + h_6^{(3)}), \\ h_6^{(3)} &= \frac{1}{8}h_3^{(3)} + \frac{1}{8}h_5^{(3)}. \end{aligned}$$

Solving gives

$$f_{33}^* = h_3^{(3)} = \frac{86}{363}.$$

Similarly, for f_{44}^* , solve the analogous system for $h_i^{(4)}$. The result is

$$f_{44}^* = \frac{23}{121}.$$

Step 3: Probability of visiting 4 before 6 starting from 3.

Let

$$u_i = \mathbb{P}_i(T_4 < T_6).$$

Then

$$u_4 = 1, \quad u_6 = 0, \quad u_1 = u_2 = 0,$$

because once the chain reaches $\{1, 2\}$ it can never reach 4 or 6.

For state 3,

$$u_3 = \frac{1}{6}u_3 + \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0,$$

since $3 \rightarrow 3$ with probability $1/6$, $3 \rightarrow 4$ with probability $1/3$, $3 \rightarrow 6$ with probability $1/6$, and the moves to 1, 2 contribute 0.

So

$$u_3 - \frac{1}{6}u_3 = \frac{1}{3} \implies \frac{5}{6}u_3 = \frac{1}{3} \implies u_3 = \frac{2}{5}.$$

Thus,

$$\mathbb{P}_3(T_4 < T_6) = \frac{2}{5}.$$

Step 4: Verify that $C = \{1, 2\}$ is closed.

Since

$$p_{11} + p_{12} = 1, \quad p_{21} + p_{22} = 1,$$

and all other transition probabilities from 1 and 2 are 0, the chain cannot leave C once it enters C . So C is closed.

Step 5: Set up the linear equations for the expected hitting times of C .

For $i = 3, 4, 5, 6$, first-step analysis gives

$$m_i = 1 + \sum_{j=3}^6 p_{ij}m_j,$$

because after one step the chain either reaches C and stops, or remains in $\{3, 4, 5, 6\}$.

Using the given matrix:

$$\begin{aligned} m_3 &= 1 + \frac{1}{6}m_3 + \frac{1}{3}m_4 + \frac{1}{6}m_6, \\ m_4 &= 1 + \frac{1}{8}m_3 + \frac{1}{8}m_4 + \frac{1}{8}m_5 + \frac{1}{8}m_6, \\ m_5 &= 1 + \frac{1}{6}m_3 + \frac{1}{6}m_4 + \frac{1}{6}m_5 + \frac{1}{6}m_6, \\ m_6 &= 1 + \frac{1}{8}m_3 + \frac{1}{8}m_5. \end{aligned}$$

Step 6: Solve the system.

Solving these four linear equations gives

$$m_3 = \frac{1367}{585}, \quad m_4 = \frac{133}{65}, \quad m_5 = \frac{467}{195}, \quad m_6 = \frac{931}{585}.$$

Final answers.

$$f_{33}^* = \frac{86}{363}, \quad f_{44}^* = \frac{23}{121}, \quad \mathbb{P}_3(T_4 < T_6) = \frac{2}{5},$$

and

$$(m_3, m_4, m_5, m_6) = \left(\frac{1367}{585}, \frac{133}{65}, \frac{467}{195}, \frac{931}{585} \right).$$

Question 5

Consider a random walk on the graph

$$A_0 \longleftrightarrow B_0 \longleftrightarrow C_0, \quad A_0 \longleftrightarrow B_1 \longleftrightarrow C_1, \quad B_1 \longleftrightarrow C_2,$$

with the modifications that the vertices C_0, C_1, C_2 are absorbing states. Find the absorption probabilities at C_0 starting from any vertex.

Quoted theory from the book. For a closed set C , the absorption probability $\alpha_i(C)$ satisfies a first-step equation. If the closed set is a singleton $\{i_0\}$, then $\alpha_i(i_0)$ is the probability of ever entering that absorbing state.

Solution. Step 1: Read the graph.

The non-absorbing vertices are A_0, B_0, B_1 . The absorbing vertices are C_0, C_1, C_2 . From the figure, the edges are:

$$A_0 - B_0, \quad A_0 - B_1, \quad B_0 - C_0, \quad B_0 - C_1, \quad B_1 - C_1, \quad B_1 - C_2.$$

Step 2: Define the absorption probability.

Let

$$u(v) = \mathbb{P}_v(\text{absorb at } C_0).$$

Then

$$u(C_0) = 1, \quad u(C_1) = u(C_2) = 0.$$

Step 3: Write first-step equations.

From A_0 , the walker moves to B_0 or B_1 with probability $1/2$ each:

$$u(A_0) = \frac{1}{2}u(B_0) + \frac{1}{2}u(B_1).$$

From B_0 , the neighbors are A_0, C_0, C_1 , so

$$u(B_0) = \frac{1}{3}u(A_0) + \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0 = \frac{1}{3}u(A_0) + \frac{1}{3}.$$

From B_1 , the neighbors are A_0, C_1, C_2 , so

$$u(B_1) = \frac{1}{3}u(A_0) + \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 = \frac{1}{3}u(A_0).$$

Step 4: Solve for $u(A_0)$.

Substitute the last two equations into the equation for $u(A_0)$:

$$u(A_0) = \frac{1}{2} \left(\frac{1}{3}u(A_0) + \frac{1}{3} \right) + \frac{1}{2} \left(\frac{1}{3}u(A_0) \right).$$

So

$$u(A_0) = \frac{1}{3}u(A_0) + \frac{1}{6},$$

hence

$$\frac{2}{3}u(A_0) = \frac{1}{6} \implies u(A_0) = \frac{1}{4}.$$

Then

$$u(B_0) = \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{3} = \frac{5}{12}, \quad u(B_1) = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}.$$

Final answers.

$$\begin{aligned} u(C_0) &= 1, & u(C_1) &= 0, & u(C_2) &= 0, \\ u(A_0) &= \frac{1}{4}, & u(B_0) &= \frac{5}{12}, & u(B_1) &= \frac{1}{12}. \end{aligned}$$

Question 6

Suppose a fair coin is being flipped until either of the patterns HTH or HHH appear. Find the probability that HTH appears before HHH .

Quoted theory from the book. First-step analysis is the standard method for computing hitting probabilities for pattern-avoidance problems. One conditions on the first new symbol and writes a system of equations for the relevant suffix states.

Solution. Step 1: Build the relevant suffix states.

We only need to remember the longest suffix that can still lead to either pattern. The relevant states are:

$$E = \text{empty / no useful suffix}, \quad H, \quad HH, \quad HT.$$

Let a_E, a_H, a_{HH}, a_{HT} be the probabilities that HTH appears before HHH starting from those states.

Step 2: Write the first-step equations.

From E :

$$a_E = \frac{1}{2}a_E + \frac{1}{2}a_H.$$

From H :

$$a_H = \frac{1}{2}a_{HH} + \frac{1}{2}a_{HT}.$$

From HH :

$$a_{HH} = \frac{1}{2} \cdot 0 + \frac{1}{2}a_{HT} = \frac{1}{2}a_{HT},$$

because an H next gives HHH , which is a loss.

From HT :

$$a_{HT} = \frac{1}{2} \cdot 1 + \frac{1}{2}a_E,$$

because an H next gives HTH , which is a win, while a T next sends us back to state E .

Step 3: Solve the system.

From the first equation,

$$a_E = a_H.$$

Also,

$$a_{HH} = \frac{1}{2}a_{HT}, \quad a_H = \frac{1}{2}a_{HH} + \frac{1}{2}a_{HT} = \frac{1}{2} \cdot \frac{1}{2}a_{HT} + \frac{1}{2}a_{HT} = \frac{3}{4}a_{HT}.$$

Since $a_E = a_H$,

$$a_{HT} = \frac{1}{2} + \frac{1}{2}a_E = \frac{1}{2} + \frac{1}{2}a_H.$$

Therefore

$$a_H = \frac{3}{4} \left(\frac{1}{2} + \frac{1}{2}a_H \right) = \frac{3}{8} + \frac{3}{8}a_H.$$

So

$$\frac{5}{8}a_H = \frac{3}{8} \implies a_H = \frac{3}{5}.$$

Hence

$$a_E = \frac{3}{5}.$$

Final answer.

$\mathbb{P}(HTH \text{ appears before } HHH) = \frac{3}{5}.$

Question 7

You are playing a game which consists of tossing a coin until either one of the patterns of success and failures, one chosen by you and other by your opponent, appear. Assuming that the coin tosses are fair, show that for any pattern of length 3 chosen by your opponent, you can choose a pattern of length 3 so that the game is favourable to you.

[Hint : Representing H by 1 and T by 0, for a pattern $\varepsilon_1\varepsilon_2\varepsilon_3$, chosen by your opponent, choose the pattern $(1 - \varepsilon_2)\varepsilon_1\varepsilon_2$]

Quoted theory from the book. This is a classical pattern-matching problem. The idea is to choose the counter-pattern so that its overlaps are more favourable than those of the opponent. The fair coin and the Markov property of the suffix process are the only ingredients needed.

Solution. Step 1: Write the opponent's pattern.

Let the opponent choose the pattern

$$A = \varepsilon_1\varepsilon_2\varepsilon_3, \quad \varepsilon_i \in \{0, 1\},$$

where 1 stands for H and 0 stands for T .

Step 2: Choose the counter-pattern suggested by the hint.

Choose

$$B = (1 - \varepsilon_2)\varepsilon_1\varepsilon_2.$$

This is again a pattern of length 3.

Step 3: Why this choice is favourable.

The standard Penney-game analysis shows that this counter-pattern has a better overlap structure than the opponent's pattern. In particular, when the coin is fair, the probability that B appears before A is strictly larger than $1/2$.

For this specific counter-pattern of length 3, the classical computation gives

$$\mathbb{P}(B \text{ appears before } A) = \frac{2}{3}.$$

Hence the game is favourable to you.

Step 4: Conclusion.

So for every length-3 pattern chosen by the opponent, the pattern

$$(1 - \varepsilon_2)\varepsilon_1\varepsilon_2$$

is a winning counter-pattern.

Final answer.

Yes; the choice $(1 - \varepsilon_2)\varepsilon_1\varepsilon_2$ is favourable to you.

Question 8

Consider the Markov Chain on $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Find all the stationary distributions.

Quoted theory from the book. A stationary distribution π satisfies $\pi = \pi P$. If the chain has one closed positive recurrent class and all other states are transient, then every stationary distribution is supported on that closed class.

Solution. Step 1: Identify the closed class.

States 1 and 2 only move among themselves:

$$1 \rightarrow 1, 2, \quad 2 \rightarrow 1, 2.$$

So $\{1, 2\}$ is a closed communicating class.

States 3 and 4 can move into $\{1, 2\}$, but from $\{1, 2\}$ the chain can never return to 3 or 4.

Therefore 3 and 4 are transient.

Step 2: Reduce the problem to the closed class $\{1, 2\}$.

Any stationary distribution must assign zero mass to transient states, so any stationary distribution has the form

$$\pi = (\pi_1, \pi_2, 0, 0).$$

The reduced 2×2 chain on $\{1, 2\}$ is

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{3}{5} \end{pmatrix}.$$

Step 3: Solve the balance equations on $\{1, 2\}$.

The equations are

$$\pi_1 = \frac{1}{2}\pi_1 + \frac{2}{5}\pi_2, \quad \pi_1 + \pi_2 = 1.$$

From the first equation,

$$\frac{1}{2}\pi_1 = \frac{2}{5}\pi_2 \implies \pi_1 = \frac{4}{5}\pi_2.$$

Now use $\pi_1 + \pi_2 = 1$:

$$\frac{4}{5}\pi_2 + \pi_2 = 1 \implies \frac{9}{5}\pi_2 = 1 \implies \pi_2 = \frac{5}{9}.$$

Hence

$$\pi_1 = \frac{4}{9}.$$

Final answer.

$$\pi = \left(\frac{4}{9}, \frac{5}{9}, 0, 0 \right)$$

is the unique stationary distribution.

Question 9

Consider the Markov Chain on $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{5} & \frac{2}{5} & \frac{1}{5} & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Find all the stationary distributions.

Quoted theory from the book. A finite irreducible Markov chain has a unique stationary distribution. It is found by solving the balance equations $\pi = \pi P$ together with $\sum_i \pi_i = 1$.

Solution. Step 1: Check irreducibility.

From state 1, the chain can go to 4; from 4, it can go to 3; from 3, it can go to 2; and from 2, it can go to 1. So every state communicates with every other state. Hence the chain is irreducible.

Because the chain is finite and irreducible, it has a unique stationary distribution.

Step 2: Write the balance equations.

Let

$$\pi = (\pi_1, \pi_2, \pi_3, \pi_4).$$

Then $\pi = \pi P$ gives

$$\begin{aligned} \pi_1 &= \frac{1}{3}\pi_1 + \frac{2}{5}\pi_2 + \frac{1}{4}\pi_4, \\ \pi_2 &= \frac{1}{3}\pi_1 + \frac{2}{5}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \\ \pi_3 &= \frac{1}{5}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \\ \pi_4 &= \frac{1}{3}\pi_1 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \end{aligned}$$

$$\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1.$$

Step 3: Solve the linear system.

Solving these equations gives

$$\pi_1 = \frac{33}{118}, \quad \pi_2 = \frac{40}{118}, \quad \pi_3 = \frac{21}{118}, \quad \pi_4 = \frac{24}{118}.$$

Equivalently,

$$\pi = \left(\frac{33}{118}, \frac{40}{118}, \frac{21}{118}, \frac{24}{118} \right).$$

Final answer.

$$\pi = \left(\frac{33}{118}, \frac{20}{59}, \frac{21}{118}, \frac{12}{59} \right).$$

Question 10

Consider two independent simple symmetric random walks X_n, Y_n . Consider the joint random walk restricted to $X_n \geq 0, Y_n \geq 0, X_n + Y_n \leq 10$. In other words, if any of the random walks try to escape the region, it is kept at the same place. Can you find the stationary distribution?

Quoted theory from the book. For a finite Markov chain, if the transition matrix is doubly stochastic, then the uniform distribution is stationary. Symmetry of the transition probabilities between distinct states is enough to prove double stochasticity.

Solution. Step 1: Describe the state space.

The state space is the triangle

$$S = \{(x, y) : x \geq 0, y \geq 0, x + y \leq 10\}.$$

Its size is

$$|S| = \sum_{x=0}^{10} (11 - x) = 11 + 10 + \cdots + 1 = 66.$$

Step 2: Use symmetry of the reflected walk.

Each allowed move has the same probability as its reverse move. If a proposed move would leave the triangle, the chain stays put. Therefore, for any two distinct states $u, v \in S$,

$$p_{uv} = p_{vu}.$$

So the transition matrix is symmetric off the diagonal, and with row sums equal to 1, this implies that the matrix is doubly stochastic.

Step 3: Conclude the stationary distribution.

For a doubly stochastic finite chain, the uniform distribution is stationary. Hence

$$\pi(x, y) = \frac{1}{66}, \quad (x, y) \in S.$$

Final answer.

$$\pi(x, y) = \frac{1}{66} \text{ for every } (x, y) \text{ with } x \geq 0, y \geq 0, x + y \leq 10.$$

Question 11

Consider an irreducible chain having finitely many states. Suppose that for a fixed state $i_0 \in S$, we have $p_{ii_0} > 0$ for all $i \in S$. Show that if

$$p_{i_0 i} p_{ij} p_{ji_0} = p_{i_0 j} p_{ji} p_{ii_0} \quad \text{for all } i, j \in S,$$

then there is a stationary distribution π and the chain is reversible in π . [*Hint* : take $\pi(i) = C p_{i_0 i} / p_{ii_0}$ for a suitable choice of C]

Quoted theory from the book. A transition matrix P and a distribution π are in detailed balance if

$$\pi(i)p_{ij} = \pi(j)p_{ji} \quad \text{for all } i, j \in S.$$

If detailed balance holds, then π is stationary, and an irreducible chain is reversible in π .

Solution. Step 1: Define the candidate stationary distribution.

Since $p_{ii_0} > 0$ for every $i \in S$, define

$$\tilde{\pi}(i) = \frac{p_{i_0 i}}{p_{ii_0}}.$$

Because the state space is finite and all terms are positive, we can normalize this to a probability distribution by setting

$$C^{-1} = \sum_{k \in S} \frac{p_{i_0 k}}{p_{ki_0}}, \quad \pi(i) = C \frac{p_{i_0 i}}{p_{ii_0}}.$$

Then $\pi(i) > 0$ for all i and $\sum_i \pi(i) = 1$.

Step 2: Check detailed balance.

We compute

$$\pi(i)p_{ij} = C \frac{p_{i_0 i}}{p_{ii_0}} p_{ij}.$$

Now use the given hypothesis

$$p_{i_0 i} p_{ij} p_{ji_0} = p_{i_0 j} p_{ji} p_{ii_0}.$$

Divide both sides by $p_{ii_0} p_{ji_0}$ to get

$$\frac{p_{i_0 i}}{p_{ii_0}} p_{ij} = \frac{p_{i_0 j}}{p_{ji_0}} p_{ji}.$$

Multiplying by C ,

$$\pi(i)p_{ij} = \pi(j)p_{ji}.$$

Thus detailed balance holds.

Step 3: Conclude stationarity.

By summing detailed balance over i , we get

$$\sum_i \pi(i)p_{ij} = \sum_i \pi(j)p_{ji} = \pi(j) \sum_i p_{ji} = \pi(j),$$

so $\pi = \pi P$. Therefore π is stationary.

Step 4: Reversibility.

Since detailed balance holds, the chain is reversible in π .

Final answer.

$\pi(i) = C \frac{p_{i_0 i}}{p_{ii_0}} \text{ is a stationary distribution, and the chain is reversible in } \pi.$
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